ANDROID DIGITAL FORENSICS EXAMINATION REPORT

SBT-DF205 — Case Study 1

Evidence source: Pixel 3.zip
Examiner: Charles Uke
Environment: Kali Linux Rolling 2025.3
Report date: 26 October 2026

Student id : 2025/FWSD/11466

Prepared from recorded terminal outputs and case notes. Windows screenshots were not directly available for inspection during report preparation.

Document Control

Field	Details
Case reference	SBT-DF205-C1
Case title	Android Digital Forensics Examination — Pixel 3
Examiner	Charles Uke
Source archive	/home/kali/Downloads/Pixel 3.zip
Preserved copy	original/Pixel 3.zip
Extracted evidence	working/extracted/Pixel 3/
Working databases	working/databases/
Primary outputs	outputs/
Notes	notes/
Screenshot storage	Separate Windows folder; not found under the Kali case directory during inventory.

Executive Summary

This examination reviewed a supplied Android evidence archive named Pixel 3.zip. The archive was preserved as a case copy, its SHA-256 hash was recorded, and the ZIP integrity test reported no errors. The inventory reported 76,403 entries, including 64,550 files, and 8,991,963,938 bytes of uncompressed content. Extraction verification reported 64,550 files.

The extracted artifacts identify a Google Pixel 3 / blueline software build running Android 10, build QQ1A.200105.003, security patch 2020-01-01. Two Android user profiles were found in system user records. Selected settings showed Wi-Fi enabled, Bluetooth disabled, location mode configured for GPS/network providers, and ADB enabled.

Communications examination covered the telephony database, call log, contacts provider, and calendar database. The telephony database contained 24 SMS records, 7 MMS metadata records, 10 MMS parts, and 14 MMS address rows. The call-log database contained 14 records. The calendar database contained one calendar row but no event rows. The contacts provider contained one aggregate contact, ten raw-contact rows, 35 data rows, and 30 MIME types.

A combined communications timeline was created and validated. It contains 45 rows: 14 CALL, 24 SMS, and 7 MMS. Required columns were present, timestamps were chronologically sorted, and the timeline spans 2020-01-29 18:20:45 UTC to 2020-02-14 14:41:18 UTC. The timeline SHA-256 is f5a2ca029e8a0ab82b31319f644dba16dbb6fc7eee33a35441ed0dfcda5af8c8.

Stored records correlate communications with number variants around 9195790479 / +19195790479 on 9 and 14 February 2020. These are database correlations only. They do not independently establish who physically used the device, the real-world identity of a contact, or who authored a message.

1. Objectives and Scope

Objectives were to document and preserve the supplied archive, verify integrity, characterize the extracted Android environment, identify relevant communication databases, summarize SMS/MMS/call/contact/calendar artifacts, correlate selected records by number and timestamp, and produce a unified timeline.

This report is based on recorded outputs and results provided during the examination session. It does not claim exhaustive review of all 64,550 extracted files, independent acquisition from a physical handset, or direct inspection of the Windows screenshot files.

2. Evidence Handling and Integrity

2.1 Archive and extraction

Item	Recorded result
Archive	Pixel 3.zip; 5,247,820,897 bytes
Preserved archive SHA-256	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a
Archive integrity	unzip -t reported no errors

Archive inventory	76,403 entries; 64,550 files; 8,991,963,938 bytes uncompressed
Extraction path	working/extracted/Pixel 3/; 64,550 files
Extraction start	2026-10-26 15:10:17 UTC
Extraction verification	2026-10-26 15:28:14 UTC
Original receipt time	Not established in the handling notes

The recorded source and preserved archive hashes matched. The notes include archive_source_sha256.txt, archive_integrity_check.txt, extraction_times.txt, initial_environment.txt, and Handling_Log.csv.

2.2 Working database hashes

Database	Recorded SHA-256
mmssms.db	cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342
contacts2.db	2a1fee3d9412e2f746592fcc0f7abcdd0ece26bc02c95240fd210464f4625050
calllog.db	5979a3fa0b4e70aff6dc20675e84a14fa584e52ad4d224d9da14d8aa0ef809d3
calendar.db	82a580901fca7580dd8841a1c988dcaa65afef068f386d603aeec6039e5aca3c

The case notes state that source and working copy hashes matched for these databases. Database examination was conducted using working copies where documented.

2.3 Environment

Recorded tools/environment: Kali Linux Rolling 2025.3; Python 3.14.7; SQLite 3.46.1; coreutils sha256sum 9.7; unzip 6.00; ExifTool 13.25.

3. Device and User-Profile Findings

3.1 Build properties

Property	Observed value
Android version	10
Build ID	QQ1A.200105.003
Security patch	2020-01-01
Fingerprint	google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
Device name setting	Pixel 3

Source: working/extracted/Pixel 3/system/build.prop. These artifacts identify stored build properties; they do not independently establish the physical handset's provenance.

3.2 User profiles

User ID	Stored profile label	Last login (UTC)
0	This Is	2020-02-14 15:51:57.124
11	User 2	2020-02-14 12:50:01.922

The user list record reported nextSerialNumber 12, version 7, user IDs 0 and 11, and deviceOwnerUserId -10000. Guest restrictions included no_sms, no_install_unknown_sources, no_config_wifi, and no_outgoing_calls. Stored profile names are not verified real world identities.

3.3 Selected settings

Source	Recorded settings	Interpretive limit
settings_global.xml	development_settings_enabled=1; adb_enabled=1; airplane_mode_on=0; wifi_on=1; bluetooth_on=0; device_name=Pixel 3	Configuration values do not provide a complete history of actual connectivity.
settings_secure.xml	location_mode=3; providers gps,network; user_setup_complete=1; mock_location=0	Does not prove a location fix or sharing event.

settings_system.xml	screen_off_timeout=300000 ms	Stored configuration only.

Android ID and Bluetooth address were present in source output but are omitted to minimize exposure of device identifiers.

4. SMS and MMS Examination

4.1 Telephony database

Source: working/extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db. Working copy: working/databases/mmssms.db. Recorded SHA-256: cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342. SQLite integrity_check returned 'ok'.

Table	Count
sms	24
pdu (MMS metadata)	7
part (MMS parts)	10
addr (MMS addresses)	14

Schema/counts and detail outputs include outputs/mmssms_schema.txt, mmssms_counts.txt, sms_summary.txt, sms_details.txt, mms_summary.txt, mms_parts_and_addresses.txt, and mms_text_parts.txt.

4.2 SMS findings

SMS IDs 2–25 span 2020-01-29 18:20:45 UTC to 2020-02-14 13:09:03 UTC. Records include service generated verification notifications and account alerts associated with several applications. One time verification codes and sensitive URLs are not reproduced.

SMS ID	UTC	Stored type	Observed content / note
21	2020-02-09 18:41:42	2	“I am on my way to the coffee shop. Do you want anything?” Address 9195790479; 56 characters.
25	2020-02-14 13:09:03	2	“Wow, the messages sync over between accounts.” Address 919-579-0479; 46 characters.

Type codes are reported as stored. Directional labels such as sent/received were not assigned without explicit validation of the applicable schema conventions.

4.3 MMS findings

MMS ID	UTC timestamp	msg_box	m_type	Text part
1	2020-02-01 20:04:53	2	128	Here's Signal.
2	2020-02-01 20:06:04	1	132	It looks like the messaging is run through the Messages app. Strange.
3	2020-02-01 20:07:52	2	128	It might be an attempt at unified messaging...?
4	2020-02-01 20:08:12	1	132	Maybe?
5	2020-02-01 20:09:27	2	128	It will be interesting to see where the messages are stored.
6	2020-02-09 18:43:08	1	132	No, thank you
7	2020-02-14 13:10:08	1	132	That is interesting.

Address rows contain number variants 9195790479 and +19195790479. MMS IDs 1, 3, and 5 included grouped addresses +19195794674 and 9195790479. Some incoming looking records had duplicate address rows with stored address types 137 and 151. These codes were not interpreted without validation.

5. Call-Log Examination

Source: working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db. Working copy: working/databases/calllog.db. SHA-256: 5979a3fa0b4e70aff6dc20675e84a14fa584e52ad4d224d9da14d8aa0ef809d3. Integrity_check returned 'ok'. Fourteen rows were recorded.

Call ID(s)	Observed result
1–5	Type code 3; duration 0; no displayed number.
6	2020-02-08 20:14:56 UTC; type 3; duration 0; +332629243087.
7–8	2020-02-12 15:37:29 and 15:38:21 UTC; +19195121037; type codes 3 and 4; second duration 3 seconds.
9–10	2020-02-13 00:52:44 and 00:53:33 UTC; +15755863247; type codes 3 and 4; second duration 3 seconds.
11	2020-02-14 01:29:01 UTC; type 2; 11 seconds; 9195790479; stored name Josh Hickman.
12	2020-02-14 01:30:24 UTC; type 1; 65 seconds; +19195790479; stored name Josh Hickman.
13	2020-02-14 13:05:57 UTC; type 2; 86 seconds; 9195790479; stored name Josh Hickman.
14	2020-02-14 14:41:18 UTC; type 1; 91 seconds; 9195790479; stored name Josh Hickman.

Call type codes remain raw numeric values because their meanings were not independently validated. Blank participant numbers were retained as blank rather than inferred.

6. Contacts and Calendar

6.1 Contacts provider

Source: working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db; working copy: working/databases/contacts2.db. SHA-256: 2a1fee3d9412e2f746592fcc0f7abcdd0ece26bc02c95240fd210464f4625050.

Table/artifact	Count
contacts	1
raw_contacts	10
data	35
mimetypes	30

Selected records display Josh Hickman and number variants including (919) 579-0479, 9195790479, and +19195790479. Raw contact IDs 6 and 7 lacked contact_id in the selected output and contained +990841016803084; the value was not validated as a valid telephone number.

App-specific MIME types included Telegram, Silent Circle, Signal, Skype, Viber, WhatsApp, and Google Tachyon categories. Such stored provider metadata is not proof of active use, message exchange, or account ownership.

A contacts2.db integrity_check attempt returned 'no such collation sequence: PHONEBOOK'. Subsequent basic queries succeeded. This is consistent with the SQLite CLI lacking an Android-specific collation; it is not proof of corruption, and a successful full integrity check is not claimed.

6.2 Calendar database

Source: working/extracted/Pixel 3/data/data/com.android.providers.calendar/databases/calendar.db; working copy: working/databases/calendar.db. SHA-256: 82a580901fca7580dd8841a1c988dcaa65afef068f386d603aeec6039e5aca3c. Integrity_check returned 'ok'.

Calendar table	Count
Events	0
Calendars	1
Attendees	0
Reminders	0

The single calendar row was a Google calendar with visible=1, sync_events=1, isPrimary=1, deleted=0. Its account identifier is omitted. No event rows were returned; this does not rule out deleted, unsynchronized, or external events.

7. Unified Communications Timeline

The combined CSV is outputs/communications_timeline.csv. It combines call-log, SMS, and MMS records, normalizes their stored dates into UTC-formatted timestamps, and sorts by timestamp, artifact type, and record ID.

Validation	Verified result
Rows	45
CALL	14
SMS	24
MMS	7
Required columns	All seven present
Chronology	True
First event	2020-01-29 18:20:45 UTC
Last event	2020-02-14 14:41:18 UTC
SHA-256	f5a2ca029e8a0ab82b31319f644dba16dbb6fc7eee33a35441ed0dfcda5af8c8

7.1 Correlation observations

- On 2020-02-09, SMS ID 21 is timestamped 18:41:42 UTC and uses address 9195790479. MMS ID 6 follows at 18:43:08 UTC and contains “No, thank you”. This is a temporal/number correlation, not independent proof of message direction.
- On 2020-02-14, call ID 13 is timestamped 13:05:57 UTC and uses 9195790479. SMS ID 25 follows at 13:09:03 UTC with formatted number 919-579-0479. MMS ID 7 follows at 13:10:08 UTC and uses +19195790479. Call ID 14 is timestamped 14:41:18 UTC.
- The contacts provider associates number variants with the stored label Josh Hickman; this is not independent identity verification.
- MMS IDs 1, 3, and 5 on 1 February contain text referring to Signal and where messages are stored. Their content does not prove a particular app carried the surrounding communications.

The timeline type_code column combines different table specific fields: call-log type, SMS type, and MMS msg_box. They are not directly comparable. content_length is a text character count for SMS and the sum of available text/plain MMS part character counts, not a universal byte-size measure.

8. Commands and Queries

The following are commands documented in the session or representative forms consistent with the recorded workflow. Because a complete shell history export was not supplied, this is not asserted to be an exhaustive verbatim transcript of every command.

8.1 Archive checks

```
sha256sum original/'Pixel 3.zip'
unzip -t original/'Pixel 3.zip'
find working/extracted/'Pixel 3' -type f | wc -l
```

8.2 Database integrity and hashes

```
sha256sum working/extracted/Pixel\ 3/data/data/com.android.providers.telephony/databases/mmssms.db \
          working/databases/mmssms.db
sqlite3 working/databases/mmssms.db 'PRAGMA integrity_check;'
sqlite3 working/databases/calllog.db 'PRAGMA integrity_check;'
sqlite3 working/databases/calendar.db 'PRAGMA integrity_check;'
```

8.3 Timeline generation SQL

```
cd ~/SBT-DF205-Case1
sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/communications_timeline.csv
.headers on
.mode csv
ATTACH DATABASE 'working/databases/calllog.db' AS call_db;
SELECT event_utc, artifact_type, record_id, participant, type_code,
       duration_seconds, content_length
FROM (
 SELECT datetime(date/1000,'unixepoch') AS event_utc, 'CALL' AS artifact_type,
        _id AS record_id, number AS participant, type AS type_code,
        duration AS duration_seconds, NULL AS content_length
 FROM call_db.calls
 UNION ALL
 SELECT datetime(date/1000,'unixepoch'), 'SMS', _id, address, type,
        NULL, length(body)
 FROM sms
 UNION ALL
 SELECT datetime(p.date,'unixepoch'), 'MMS', p._id,
        (SELECT group_concat(DISTINCT a.address) FROM addr a WHERE a.msg_id=p._id),
        p.msg_box, NULL,
        (SELECT SUM(length(pt.text)) FROM part pt
         WHERE pt.mid=p._id AND pt.ct='text/plain')
 FROM pdu p
)
ORDER BY event_utc, artifact_type, record_id;
SQL
```

8.4 Timeline validation script

```
python3 - <<'PY'
import csv, hashlib
from collections import Counter
from pathlib import Path
from datetime import datetime

path = Path("outputs/communications_timeline.csv")
raw = path.read_bytes()
rows = list(csv.DictReader(path.open(newline="", encoding="utf-8")))
required = {"event_utc","artifact_type","record_id","participant",
            "type_code","duration_seconds","content_length"}
print("SHA-256:", hashlib.sha256(raw).hexdigest())
print("Data rows:", len(rows))
print("Required columns present:", required.issubset(rows[0].keys()) if rows else False)
print("Counts:", Counter(row["artifact_type"] for row in rows))
times = [datetime.strptime(row["event_utc"], "%Y-%m-%d %H:%M:%S") for row in rows]
print("Chronologically sorted:", times == sorted(times))
print("First event:", rows[0]["event_utc"] if rows else "N/A")
print("Last event:", rows[-1]["event_utc"] if rows else "N/A")
PY
```

8.5 Hash and case inventory

```
sha256sum outputs/communications_timeline.csv | tee outputs/communications_timeline_sha256.txt
tail -n 5 notes/Handling_Log.csv
find outputs -maxdepth 1 -type f -printf '%f (%s bytes)\\n' | sort
find notes -maxdepth 1 -type f -printf '%f (%s bytes)\\n' | sort
find . -type f \\( -iname 'SS-*.png' -o -iname 'SS-*.jpg' \\) -printf '%p\\n' | sort
```

9. Limitations

- The examination begins with a supplied ZIP archive; a physical handset acquisition and independent device to archive provenance record are not documented in the available notes.
- The archive receipt time was not established.
- The report relies on recorded terminal results and notes. Figures should be compared with the current local outputs before submission.
- Windows Snipping Tool screenshots were stored separately and were not present under the Kali case directory during inventory. Their content has not been independently inspected for this report.
- outputs/extraction.log was zero bytes. Extraction was separately reported as verified by file counts/timing notes; the empty log should be disclosed.
- The contacts database integrity check did not complete due to missing PHONEBOOK collation. Basic queries succeeded, but no full integrity check success is claimed.
- Call/SMS/MMS numeric type fields are preserved without directional interpretation where their meanings were not validated.
- Contact labels and telephone number variants are stored associations, not independently verified identities.
- App specific MIME types do not prove active application use. Zero calendar event rows do not rule out deleted, unsynchronized, or external events.
- Verification codes and sensitive URLs were intentionally excluded. The full extracted evidence was not exhaustively reviewed.

10. Conclusion

The supplied archive passed the recorded ZIP integrity test and its SHA-256 was documented. The extracted Android artifacts identify Android 10 on a Google Pixel 3 / blueline build. The telephony, call-log, contacts, and calendar databases were located and examined using working copies, and selected hashes, schemas, counts, and correlations were recorded.

The principal derived artifact is a validated 45 row communications timeline covering 29 January through 14 February 2020. It provides a reproducible basis for comparing call, SMS, and MMS records. The records show temporal associations involving number variants around 9195790479 / +19195790479, and the contacts provider links number variants to the stored label Josh Hickman. These findings do not independently establish real world identity, message direction, or who operated the device.

11. Screenshot Reference Register

These are proposed reference names for screenshots taken with Windows Snipping Tool. The Kali inventory showed no SS-*.png or SS-*.jpg files in the case folder at that time, so the list is a checklist, not confirmation that every image exists. Match each reference to the actual Windows filename before submission.

Reference	Content to show
SS-01_Case_Directory.png	Case folder structure
SS-02_Tool_Versions.png	Kali and tool versions
SS-03_Archive_Integrity.png	Archive hash and ZIP integrity
SS-04_Extraction_Verification.png	Extraction count and verification
SS-05_Device_Properties.png	Android version/build/patch
SS-06_User_Profiles.png	User IDs and login times
SS-07_Settings_Connectivity.png	Selected settings; redact identifiers
SS-08_SMS_Database.png	SMS schema/counts/hash/integrity
SS-09_SMS_Timeline.png	Selected SMS IDs/timestamps; redact codes
SS-11_MMS_Summary.png	MMS counts and timestamps
SS-14_Contact_Call_Databases.png	Database locations
SS-18_Call_Log_Summary.png	Call-log summary
SS-20_Calendar_Records.png	Calendar counts and zero events
SS-23_Contacts_Summary.png	Contacts counts/MIME types
SS-28_Contact_Call_Correlation.png	Contact-to-call correlation
SS-29_Contact_SMS_Correlation.png	Contact-to-SMS correlation
SS-31_Combined_Communications_Timeline.png	45-row timeline
SS-32_Timeline_Validation.png	Columns/counts/chronology/hash

SS-33_Timeline_Evidence_Log.png	Timeline hash and handling log

Appendix A — Key File Locations

Purpose	Path
Case root	~/SBT-DF205-Case1/
Preserved archive	original/Pixel 3.zip
Extracted evidence	working/extracted/Pixel 3/
Working databases	working/databases/
Combined timeline	outputs/communications_timeline.csv
Timeline validation	outputs/timeline_validation.txt
Timeline hash	outputs/communications_timeline_sha256.txt
Archive hash	notes/archive_source_sha256.txt
Handling log	notes/Handling_Log.csv
Device findings	notes/device_profile_findings.txt
Database inventory	outputs/database_inventory.txt
SMS details	outputs/sms_details.txt
MMS details	outputs/mms_parts_and_addresses.txt; outputs/mms_text_parts.txt
Call summary	outputs/calllog_summary.txt
Contacts	outputs/contacts_selected_details.txt; outputs/contacts_app_specific_details.txt
Correlation outputs	outputs/contact_call_correlation.txt; outputs/contact_sms_correlation.txt; outputs/contact_mms_correlation.txt

Appendix B — Pre-submission Checklist

- Compare every report figure with the current case outputs on Kali.
- Transfer Windows screenshots into the submission package or submit them in a separately named folder; make report references match actual filenames.
- Redact verification codes, sensitive URLs, Android ID, Bluetooth address, and personal account identifiers where not required.
- Review the handling log for accurate dates/times and action descriptions.
- Include the communications timeline, validation output, and timeline hash if permitted by the assignment.
- Do not submit the original archive or full extracted databases unless the assignment explicitly requires them.

```
"2020-02-03 19:32:22",CALL,5,"",3,0,
"2020-02-05 16:22:21",SMS,18,32665,1,,82
"2020-02-07 08:15:59",SMS,19,32665,1,,83
"2020-02-08 18:17:01",SMS,20,32665,1,,82
"2020-02-08 20:14:56",CALL,6,+332629243087,3,0,
"2020-02-09 18:41:42",SMS,21,9195790479,2,,56
"2020-02-09 18:43:08",MMS,6,+19195790479,1,,13
"2020-02-11 18:02:40",SMS,22,32665,1,,84
"2020-02-12 15:37:29",CALL,7,+19195121037,3,0,
"2020-02-12 15:38:21",CALL,8,+19195121037,4,3,
"2020-02-12 19:12:18",SMS,23,32665,1,,83
"2020-02-13 00:52:44",CALL,9,+15755863247,3,0,
"2020-02-13 00:53:33",CALL,10,+15755863247,4,3,
"2020-02-13 21:12:24",SMS,24,32665,1,,84
"2020-02-14 01:29:01",CALL,11,9195790479,2,11,
"2020-02-14 01:30:24",CALL,12,+19195790479,1,65,
"2020-02-14 13:05:57",CALL,13,9195790479,2,86,
"2020-02-14 13:09:03",SMS,25,919-579-0479,2,,46
"2020-02-14 13:10:08",MMS,7,+19195790479,1,,20
"2020-02-14 14:41:18",CALL,14,9195790479,1,91,
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "═══ TIMELINE ROW COUNT ═══"
tail -n +2 outputs/communications_timeline.csv | wc -l

echo
echo "═══ FIRST 15 TIMELINE ROWS ═══"
head -n 16 outputs/communications_timeline.csv

echo
echo "═══ LAST 15 TIMELINE ROWS ═══"
tail -n 15 outputs/communications_timeline.csv
═══ TIMELINE ROW COUNT ═══
45

═══ FIRST 15 TIMELINE ROWS ═══
event_utc,artifact_type,record_id,participant,type_code,duration_seconds,content_length
"2020-01-29 18:20:45",SMS,2,+13233205525,1,,70
"2020-01-29 18:30:46",SMS,3,732873,1,,87
"2020-01-29 18:31:36",SMS,4,+18437514985,1,,55
"2020-01-29 18:32:08",SMS,5,+12253690855,1,,37
"2020-01-29 18:33:40",SMS,6,+12347570651,1,,37
"2020-01-29 18:35:03",SMS,7,+12349008124,1,,37
"2020-01-29 18:37:08",SMS,8,47543,1,,153
"2020-01-29 18:41:38",SMS,9,732873,1,,43
"2020-01-29 18:43:08",SMS,10,59109,1,,54
"2020-01-29 18:50:13",SMS,11,837402,1,,47
"2020-01-29 18:50:24",SMS,12,32665,1,,194
"2020-01-29 20:16:10",SMS,13,732873,1,,55
"2020-01-29 21:15:33",SMS,14,44398,1,,66
```

```
"2020-01-29 18:32:08",SMS,5,+12253690855,1,,37
"2020-01-29 18:33:40",SMS,6,+12347570651,1,,37
"2020-01-29 18:35:03",SMS,7,+12349008124,1,,37
"2020-01-29 18:37:08",SMS,8,47543,1,,153
"2020-01-29 18:41:38",SMS,9,732873,1,,43
"2020-01-29 18:43:08",SMS,10,59109,1,,54
"2020-01-29 18:50:13",SMS,11,837402,1,,47
"2020-01-29 18:50:24",SMS,12,32665,1,,194
"2020-01-29 20:16:10",SMS,13,732873,1,,55
"2020-01-29 21:15:33",SMS,14,44398,1,,66
"2020-01-31 15:16:36",CALL,1,"",3,0,
"2020-01-31 18:07:50",CALL,2,"",3,0,

═══ LAST 15 TIMELINE ROWS ═══
"2020-02-09 18:41:42",SMS,21,9195790479,2,,56
"2020-02-09 18:43:08",MMS,6,+19195790479,1,,13
"2020-02-11 18:02:40",SMS,22,32665,1,,84
"2020-02-12 15:37:29",CALL,7,+19195121037,3,0,
"2020-02-12 15:38:21",CALL,8,+19195121037,4,3,
"2020-02-12 19:12:18",SMS,23,32665,1,,83
"2020-02-13 00:52:44",CALL,9,+15755863247,3,0,
"2020-02-13 00:53:33",CALL,10,+15755863247,4,3,
"2020-02-13 21:12:24",SMS,24,32665,1,,84
"2020-02-14 01:29:01",CALL,11,9195790479,2,11,
"2020-02-14 01:30:24",CALL,12,+19195790479,1,65,
"2020-02-14 13:05:57",CALL,13,9195790479,2,86,
"2020-02-14 13:09:03",SMS,25,919-579-0479,2,,46
"2020-02-14 13:10:08",MMS,7,+19195790479,1,,20
"2020-02-14 14:41:18",CALL,14,9195790479,1,91,

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ >....
from datetime import datetime

path = Path("outputs/communications_timeline.csv")
raw = path.read_bytes()
rows = list(csv.DictReader(path.open(newline="", encoding="utf-8")))

required = {
    "event_utc", "artifact_type", "record_id", "participant",
    "type_code", "duration_seconds", "content_length"
}

print("===== TIMELINE VALIDATION =====")
print(f"File: {path}")
print(f"SHA-256: {hashlib.sha256(raw).hexdigest()}")
print(f"Data rows: {len(rows)}")
print(f"Required columns present: {required.issubset(rows[0].keys()) if rows else False}")

counts = Counter(row["artifact_type"] for row in rows)
print("\n===== COUNTS BY ARTIFACT TYPE =====")
for kind in sorted(counts):
    print(f"{kind}: {counts[kind]}")

times = [
    datetime.strptime(row["event_utc"], "%Y-%m-%d %H:%M:%S")
    for row in rows
]
print("\n===== CHRONOLOGY CHECK =====")
print(f"Chronologically sorted: {times == sorted(times)}")
print(f"First event (UTC): {rows[0]['event_utc'] if rows else 'N/A'}")
```

```
counts = Counter(row["artifact_type"] for row in rows)
print("\n===== COUNTS BY ARTIFACT TYPE =====")
for kind in sorted(counts):
    print(f"{kind}: {counts[kind]}")

times = [
    datetime.strptime(row["event_utc"], "%Y-%m-%d %H:%M:%S")
    for row in rows
]
print("\n===== CHRONOLOGY CHECK =====")
print(f"Chronologically sorted: {times == sorted(times)}")
print(f"First event (UTC): {rows[0]['event_utc'] if rows else 'N/A'}")
print(f"Last event (UTC): {rows[-1]['event_utc'] if rows else 'N/A'}")
PY
===== TIMELINE VALIDATION =====
File: outputs/communications_timeline.csv
SHA-256: f5a2ca029e8a0ab82b31319f644dba16dbb6fc7eee33a35441ed0dfcda5af8c8
Data rows: 45
Required columns present: True

===== COUNTS BY ARTIFACT TYPE =====
CALL: 14
MMS: 7
SMS: 24

===== CHRONOLOGY CHECK =====
Chronologically sorted: True
First event (UTC): 2020-01-29 18:20:45
Last event (UTC): 2020-02-14 14:41:18

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "═══ CASE DIRECTORY ═══"
find . -maxdepth 2 -type d | sort

echo
echo "═══ OUTPUT FILES ═══"
find outputs -maxdepth 1 -type f -printf '%f (%s bytes)\n' | sort

echo
echo "═══ NOTES FILES ═══"
find notes -maxdepth 1 -type f -printf '%f (%s bytes)\n' | sort

echo
echo "═══ SCREENSHOT FILES ═══"
find . -type f \( -iname 'SS-*.png' -o -iname 'SS-*.jpg' \) -printf '%p\n' | sort

echo
echo "═══ EVIDENCE HASH RECORDS ═══"
find outputs -maxdepth 1 -type f -iname '*hash*' -printf '%f\n' | sort
═══ CASE DIRECTORY ═══
.
./evidence
./notes
./original
./outputs
./report
./screenshots
./working
./working/databases
```

```
==== CASE DIRECTORY ====
.
./evidence
./notes
./original
./outputs
./report
./screenshots
./working
./working/databases
./working/extracted

==== OUTPUT FILES ====
archive_listing.txt (10433406 bytes)
archive_test.txt (9516715 bytes)
calendar_account_details.txt (444 bytes)
calendar_counts.txt (300 bytes)
calendar_events.txt (0 bytes)
calendar_schema.txt (3788 bytes)
calllog_schema.txt (1276 bytes)
calllog_summary.txt (1744 bytes)
communications_timeline.csv (2156 bytes)
communications_timeline_sha256.txt (102 bytes)
contact_call_calendar_hashes.txt (753 bytes)
contact_call_calendar_paths.txt (267 bytes)
contact_call_calendar_schema.txt (1885 bytes)
contact_call_correlation.txt (330 bytes)
contact_mms_correlation.txt (1105 bytes)
contact_mms_text_correlation.txt (1392 bytes)
contacts_app_specific_details.txt (5370 bytes)
contacts_inspection.txt (303 bytes)
contacts_mimetype_summary.txt (2398 bytes)
```

```
calllog_schema.txt (1276 bytes)
calllog_summary.txt (1744 bytes)
communications_timeline.csv (2156 bytes)
communications_timeline_sha256.txt (102 bytes)
contact_call_calendar_hashes.txt (753 bytes)
contact_call_calendar_paths.txt (267 bytes)
contact_call_calendar_schema.txt (1885 bytes)
contact_call_correlation.txt (330 bytes)
contact_mms_correlation.txt (1105 bytes)
contact_mms_text_correlation.txt (1392 bytes)
contacts_app_specific_details.txt (5370 bytes)
contacts_inspection.txt (303 bytes)
contacts_mimetype_summary.txt (2398 bytes)
contact_sms_correlation.txt (232 bytes)
contacts_schema_details.txt (6458 bytes)
contacts_selected_details.txt (3260 bytes)
database_candidates.txt (8804 bytes)
database_inventory.txt (45326 bytes)
device_build_properties.txt (558 bytes)
extraction.log (0 bytes)
mms_parts_and_addresses.txt (3674 bytes)
mmssms_copy_hashes.txt (248 bytes)
mmssms_counts.txt (300 bytes)
mmssms_schema.txt (11555 bytes)
mms_summary.txt (945 bytes)
mms_text_parts.txt (1425 bytes)
profile_login_times_utc.txt (89 bytes)
settings_global_selected.txt (1891 bytes)
settings_secure_selected.txt (2067 bytes)
settings_system_selected.txt (1275 bytes)
sms_details.txt (10871 bytes)
sms_summary.txt (1976 bytes)
```

```
device_build_properties.txt (558 bytes)
extraction.log (0 bytes)
mms_parts_and_addresses.txt (3674 bytes)
mmssms_copy_hashes.txt (248 bytes)
mmssms_counts.txt (300 bytes)
mmssms_schema.txt (11555 bytes)
mms_summary.txt (945 bytes)
mms_text_parts.txt (1425 bytes)
profile_login_times_utc.txt (89 bytes)
settings_global_selected.txt (1891 bytes)
settings_secure_selected.txt (2067 bytes)
settings_system_selected.txt (1275 bytes)
sms_details.txt (10871 bytes)
sms_summary.txt (1976 bytes)
timeline_validation.txt (390 bytes)

═══ NOTES FILES ═══
archive_integrity_check.txt (186 bytes)
archive_source_sha256.txt (99 bytes)
device_profile_findings.txt (2151 bytes)
extraction_times.txt (100 bytes)
Handling_Log.csv (569 bytes)
initial_environment.txt (629 bytes)

═══ SCREENSHOT FILES ═══

═══ EVIDENCE HASH RECORDS ═══
contact_call_calendar_hashes.txt
mmssms_copy_hashes.txt

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

cat >> notes/Handling_Log.csv <<'EOF'
SBT-DF205-C1,2026-10-26,Charles Uke,Timeline validation,outputs/communications_timeline.csv,outputs/timeline_validation.txt,Python 3,"Passed"
,"45 records: 14 CALL, 24 SMS, 7 MMS; required columns present; chronological order verified; SHA-256 recorded in validation output"
EOF

echo "===== TIMELINE HASH FILE ====="
sha256sum outputs/communications_timeline.csv | tee outputs/communications_timeline_sha256.txt

echo
echo "===== HANDLING LOG LAST ENTRIES ====="
tail -n 5 notes/Handling_Log.csv

echo
echo "===== VALIDATION FILES ====="
ls -l outputs/communications_timeline.csv \
      outputs/timeline_validation.txt \
      outputs/communications_timeline_sha256.txt
===== TIMELINE HASH FILE =====
f5a2ca029e8a0ab82b31319f644dba16dbb6fc7eee33a35441ed0dfcda5af8c8  outputs/communications_timeline.csv

===== HANDLING LOG LAST ENTRIES =====
CaseReference,DateTimeUTC,Examiner,Action,SourcePath,DestinationPath,Tool,Result,Notes
SBT-DF205-C1,2026-10-26 13:49:37 UTC,Charles Uke,Environment check,/home/kali,,Kali GNU/Linux Rolling 2025.3,Recorded,"Initial environment ti
me from terminal; actual archive receipt time not yet established"
SBT-DF205-C1,2026-10-26,Charles Uke,Timeline validation,outputs/communications_timeline.csv,outputs/timeline_validation.txt,Python 3,"Passed"
,"45 records: 14 CALL, 24 SMS, 7 MMS; required columns present; chronological order verified; SHA-256 recorded in validation output"

===== VALIDATION FILES =====
```

```
===== HANDLING LOG LAST ENTRIES =====
CaseReference,DateTimeUTC,Examiner,Action,SourcePath,DestinationPath,Tool,Result,Notes
SBT-DF205-C1,2026-10-26 13:49:37 UTC,Charles Uke,Environment check,/home/kali,,Kali GNU/Linux Rolling 2025.3,Recorded,"Initial environment ti
me from terminal; actual archive receipt time not yet established"
SBT-DF205-C1,2026-10-26,Charles Uke,Timeline validation,outputs/communications_timeline.csv,outputs/timeline_validation.txt,Python 3,"Passed"
,"45 records: 14 CALL, 24 SMS, 7 MMS; required columns present; chronological order verified; SHA-256 recorded in validation output"

===== VALIDATION FILES =====
-rw-rw-r-- 1 kali kali 2156 Oct 26 20:45 outputs/communications_timeline.csv
-rw-rw-r-- 1 kali kali  102 Oct 26 20:54 outputs/communications_timeline_sha256.txt
-rw-rw-r-- 1 kali kali  390 Oct 26 20:51 outputs/timeline_validation.txt

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
5    version                  INTEGER  1        1           0
6    dirty                    INTEGER  1        0           0
7    deleted                  INTEGER  1        0           0
8    metadata_dirty           INTEGER  1        0           0
9    contact_id               INTEGER  0                    0
10   aggregation_mode         INTEGER  1        0           0
11   aggregation_needed       INTEGER  1        1           0
12   custom_ringtone          TEXT     0                    0
13   send_to_voicemail        INTEGER  1        0           0
14   x_times_contacted        INTEGER  1        0           0
15   x_last_time_contacted    INTEGER  0                    0
16   times_contacted          INTEGER  1        0           0
17   last_time_contacted      INTEGER  0                    0
18   starred                  INTEGER  1        0           0
19   pinned                   INTEGER  1        0           0
20   display_name             TEXT     0                    0
21   display_name_alt         TEXT     0                    0
22   display_name_source      INTEGER  1        0           0
23   phonetic_name            TEXT     0                    0
24   phonetic_name_style      TEXT     0                    0
25   sort_key                 TEXT     0                    0
26   phonebook_label          TEXT     0                    0
27   phonebook_bucket         INTEGER  0                    0
28   sort_key_alt             TEXT     0                    0
29   phonebook_label_alt      TEXT     0                    0
30   phonebook_bucket_alt     INTEGER  0                    0
31   name_verified            INTEGER  1        0           0
32   sync1                    TEXT     0                    0
33   sync2                    TEXT     0                    0
34   sync3                    TEXT     0                    0
35   sync4                    TEXT     0                    0
cid  name                                  type     notnull  dflt_value  pk
---  ------------------------------------  -------  -------  ----------  --
0    _id                                   INTEGER  0                    1
1    package_id                            INTEGER  0                    0
2    mimetype_id                           INTEGER  1                    0
3    raw_contact_id                        INTEGER  1                    0
4    hash_id                               TEXT     0                    0
5    is_read_only                          INTEGER  1        0           0
```

```
cid  name                                type     notnull  dflt_value  pk
---  ----------------------------------  -------  -------  ----------  --
0    _id                                 INTEGER  0                    1
1    package_id                          INTEGER  0                    0
2    mimetype_id                         INTEGER  1                    0
3    raw_contact_id                      INTEGER  1                    0
4    hash_id                             TEXT     0                    0
5    is_read_only                        INTEGER  1        0           0
6    is_primary                          INTEGER  1        0           0
7    is_super_primary                    INTEGER  1        0           0
8    data_version                        INTEGER  1        0           0
9    data1                               TEXT     0                    0
10   data2                               TEXT     0                    0
11   data3                               TEXT     0                    0
12   data4                               TEXT     0                    0
13   data5                               TEXT     0                    0
14   data6                               TEXT     0                    0
15   data7                               TEXT     0                    0
16   data8                               TEXT     0                    0
17   data9                               TEXT     0                    0
18   data10                              TEXT     0                    0
19   data11                              TEXT     0                    0
20   data12                              TEXT     0                    0
21   data13                              TEXT     0                    0
22   data14                              TEXT     0                    0
23   data15                              TEXT     0                    0
24   data_sync1                          TEXT     0                    0
25   data_sync2                          TEXT     0                    0
26   data_sync3                          TEXT     0                    0
27   data_sync4                          TEXT     0                    0
28   carrier_presence                    INTEGER  1        0           0
29   preferred_phone_account_component_name  TEXT     0                    0
30   preferred_phone_account_id          TEXT     0                    0
cid  name      type     notnull  dflt_value  pk
---  --------  -------  -------  ----------  --
0    _id       INTEGER  0                    1
1    mimetype  TEXT     1                    0

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/contacts2.db <<'SQL' | tee outputs/contacts_schema_details.txt
.headers on
.mode column

PRAGMA table_info(contacts);
PRAGMA table_info(raw_contacts);
PRAGMA table_info(data);
PRAGMA table_info(mimetypes);
SQL
cid  name                          type     notnull  dflt_value  pk
---  ----------------------------  -------  -------  ----------  --
0    _id                           INTEGER  0                    1
1    name_raw_contact_id           INTEGER  0                    0
2    photo_id                      INTEGER  0                    0
3    photo_file_id                 INTEGER  0                    0
4    custom_ringtone               TEXT     0                    0
5    send_to_voicemail             INTEGER  1        0           0
6    x_times_contacted             INTEGER  1        0           0
7    x_last_time_contacted         INTEGER  0                    0
8    times_contacted               INTEGER  1        0           0
9    last_time_contacted           INTEGER  0                    0
10   starred                       INTEGER  1        0           0
11   pinned                        INTEGER  1        0           0
12   has_phone_number              INTEGER  1        0           0
13   lookup                        TEXT     0                    0
14   status_update_id              INTEGER  0                    0
15   contact_last_updated_timestamp  INTEGER  0                    0
cid  name                      type     notnull  dflt_value  pk
---  ------------------------  -------  -------  ----------  --
0    _id                       INTEGER  0                    1
1    account_id                INTEGER  0                    0
2    sourceid                  TEXT     0                    0
3    backup_id                 TEXT     0                    0
4    raw_contact_is_read_only  INTEGER  1        0           0
5    version                   INTEGER  1        1           0
```

```
30          vnd.android.cursor.item/com.google.android.apps.tachyon.phon  1
            e.audio

18          vnd.android.cursor.item/com.silentcircle.phone                1

21          vnd.android.cursor.item/com.skype4life.phone                  1

3           vnd.android.cursor.item/nickname                              1

12          vnd.android.cursor.item/note                                  1

25          vnd.android.cursor.item/vnd.com.viber.voip.google_voice_mess  1
            age

22          vnd.android.cursor.item/vnd.com.viber.voip.viber_number_call  1

23          vnd.android.cursor.item/vnd.com.viber.voip.viber_number_mess  1
            age

24          vnd.android.cursor.item/vnd.com.viber.voip.viber_out_call_vi  1
            ber

26          vnd.android.cursor.item/vnd.com.whatsapp.profile              1

28          vnd.android.cursor.item/vnd.com.whatsapp.video.call           1

27          vnd.android.cursor.item/vnd.com.whatsapp.voip.call            1

17          vnd.android.cursor.item/vnd.org.telegram.messenger.android.p  1
            rofile

20          vnd.android.cursor.item/vnd.org.thoughtcrime.securesms.call   1

19          vnd.android.cursor.item/vnd.org.thoughtcrime.securesms.conta  1
            ct

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/contacts2.db <<'SQL' | tee outputs/contacts_mimetype_summary.txt
.headers on
.mode column

SELECT
    m._id AS mimetype_id,
    m.mimetype,
    COUNT(d._id) AS row_count
FROM mimetypes AS m
LEFT JOIN data AS d
    ON d.mimetype_id = m._id
GROUP BY m._id, m.mimetype
HAVING COUNT(d._id) > 0
ORDER BY row_count DESC, m.mimetype;
SQL
```

mimetype_id	mimetype	row_count
7	vnd.android.cursor.item/name	10
5	vnd.android.cursor.item/phone_v2	6
11	vnd.android.cursor.item/group_membership	3
29	vnd.android.cursor.item/com.google.android.apps.tachyon.phone	1
30	vnd.android.cursor.item/com.google.android.apps.tachyon.phone.audio	1
18	vnd.android.cursor.item/com.silentcircle.phone	1
21	vnd.android.cursor.item/com.skype4life.phone	1
3	vnd.android.cursor.item/nickname	1
12	vnd.android.cursor.item/note	1

```
┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/contacts2.db <<'SQL' | tee outputs/contacts_selected_details.txt
.headers on
.mode column
.width 8 10 24 48 32 8 20

SELECT
    r._id AS raw_id,
    r.contact_id,
    r.display_name,
    m.mimetype,
    d.data1 AS value,
    d.data2 AS type_code,
    d.data3 AS label
FROM raw_contacts AS r
JOIN data AS d
    ON d.raw_contact_id = r._id
JOIN mimetypes AS m
    ON m._id = d.mimetype_id
WHERE m.mimetype IN (
    'vnd.android.cursor.item/name',
    'vnd.android.cursor.item/phone_v2',
    'vnd.android.cursor.item/nickname',
    'vnd.android.cursor.item/note'
)
ORDER BY r.contact_id, r._id, m.mimetype;
SQL
raw_id    contact_id  display_name             mimetype                                  value                          type_cod
 label
```

SQL

raw_id	contact_id	display_name	mimetype	value	type_cod label
6		Josh Hickman	vnd.android.cursor.item/name	Josh Hickman	Josh Hickman
6		Josh Hickman	vnd.android.cursor.item/phone_v2	+990841016803084	
7		Josh Hickman	vnd.android.cursor.item/name	Josh Hickman	Josh Hickman
7		Josh Hickman	vnd.android.cursor.item/phone_v2	+990841016803084	
1	10	Josh Hickman	vnd.android.cursor.item/name	Josh Hickman	Josh Hickman
1	10	Josh Hickman	vnd.android.cursor.item/nickname		1
1	10	Josh Hickman	vnd.android.cursor.item/note		
1	10	Josh Hickman	vnd.android.cursor.item/phone_v2	(919) 579-0479	2
2	10	Josh Hickman	vnd.android.cursor.item/name	Josh Hickman	Josh Hickman
3	10	Josh Hickman	vnd.android.cursor.item/name	Josh Hickman	Josh Hickman
4	10	Josh Hickman	vnd.android.cursor.item/name	Josh Hickman	Josh Hickman
4	10	Josh Hickman	vnd.android.cursor.item/phone_v2	(919) 579-0479	7
5	10	Josh Hickman	vnd.android.cursor.item/name	Josh Hickman	Josh Hickman
8	10	Josh Hickman	vnd.android.cursor.item/name	Josh Hickman	Josh

```
1         10          Josh Hickman            vnd.android.cursor.item/name                Josh Hickman                  Josh
 Hickman
1         10          Josh Hickman            vnd.android.cursor.item/nickname                                          1

1         10          Josh Hickman            vnd.android.cursor.item/note

1         10          Josh Hickman            vnd.android.cursor.item/phone_v2            (919) 579-0479                2

2         10          Josh Hickman            vnd.android.cursor.item/name                Josh Hickman                  Josh
 Hickman
3         10          Josh Hickman            vnd.android.cursor.item/name                Josh Hickman                  Josh
 Hickman
4         10          Josh Hickman            vnd.android.cursor.item/name                Josh Hickman                  Josh
 Hickman
4         10          Josh Hickman            vnd.android.cursor.item/phone_v2            (919) 579-0479                7

5         10          Josh Hickman            vnd.android.cursor.item/name                Josh Hickman                  Josh
 Hickman
8         10          Josh Hickman            vnd.android.cursor.item/name                Josh Hickman                  Josh
 Hickman
9         10          Josh Hickman            vnd.android.cursor.item/name                Josh Hickman                  Josh
 Hickman
9         10          Josh Hickman            vnd.android.cursor.item/phone_v2            9195790479                    2

10        10          Josh Hickman            vnd.android.cursor.item/name                Josh Hickman                  Josh
 Hickman
10        10          Josh Hickman            vnd.android.cursor.item/phone_v2            (919) 579-0479                2


┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/contacts2.db <<'SQL' | tee outputs/contacts_app_specific_details.txt
.headers on
.mode column
.width 8 10 24 54 40 24

SELECT
    r._id AS raw_id,
    r.contact_id,
    r.display_name,
    m.mimetype,
    d.data1,
    d.data2,
    d.data3
FROM raw_contacts AS r
JOIN data AS d
    ON d.raw_contact_id = r._id
JOIN mimetypes AS m
    ON m._id = d.mimetype_id
WHERE r.contact_id = 10
  AND m.mimetype NOT IN (
      'vnd.android.cursor.item/name',
      'vnd.android.cursor.item/phone_v2',
      'vnd.android.cursor.item/nickname',
      'vnd.android.cursor.item/note'
  )
ORDER BY r._id, m.mimetype;
SQL
raw_id    contact_id  display_name             mimetype                                                  data1
```

```
raw_id    contact_id  display_name             mimetype                                                  data1
     data2                    data3
_______ __________  ________________  ____________________________________________________  ____________________________________
___  ________________________  ____________________________
1         10          Josh Hickman             vnd.android.cursor.item/group_membership                  1


2         10          Josh Hickman             vnd.android.cursor.item/vnd.org.telegram.messenger.and  1016803084
     Telegram Profile         +19195790479
                                                roid.profile


3         10          Josh Hickman             vnd.android.cursor.item/com.silentcircle.phone          tel:+1 919-579-0479
     Silent Circle            +1 919-579-0479 Silent World call

4         10          Josh Hickman             vnd.android.cursor.item/vnd.org.thoughtcrime.securesms  (919) 579-0479
     Signal                   Signal Call (919) 579-0479
                                                .call


4         10          Josh Hickman             vnd.android.cursor.item/vnd.org.thoughtcrime.securesms  (919) 579-0479
     Signal                   Message (919) 579-0479
                                                .contact


5         10          Josh Hickman             vnd.android.cursor.item/com.skype4life.phone           9195790479
                               Skype

8         10          Josh Hickman             vnd.android.cursor.item/vnd.com.viber.voip.google_voic  +19195790479
                               +19195790479
                                                e_message
```

```
8         10          Josh Hickman             vnd.android.cursor.item/vnd.com.viber.voip.viber_numbe  +19195790479
    Viber                     Free call (+19195790479)
                              r_call


8         10          Josh Hickman             vnd.android.cursor.item/vnd.com.viber.voip.viber_numbe  +19195790479
    Viber                     Free message (+19195790479)
                              r_message


8         10          Josh Hickman             vnd.android.cursor.item/vnd.com.viber.voip.viber_out_c  +19195790479
    Viber                     Viber Out call (+19195790479)
                              all_viber


9         10          Josh Hickman             vnd.android.cursor.item/vnd.com.whatsapp.profile        19195790479@s.whatsapp.net
    WhatsApp                  Message +1 (919) 579-0479

9         10          Josh Hickman             vnd.android.cursor.item/vnd.com.whatsapp.video.call     19195790479@s.whatsapp.net
    WhatsApp                  Video call +1 (919) 579-0479

9         10          Josh Hickman             vnd.android.cursor.item/vnd.com.whatsapp.voip.call      19195790479@s.whatsapp.net
    WhatsApp                  Voice call +1 (919) 579-0479

10        10          Josh Hickman             vnd.android.cursor.item/com.google.android.apps.tachyo  (919) 579-0479
    1
                              n.phone


10        10          Josh Hickman             vnd.android.cursor.item/com.google.android.apps.tachyo  (919) 579-0479
```

```
9         10          Josh Hickman             vnd.android.cursor.item/vnd.com.whatsapp.voip.call      19195790479@s.whatsapp.net
    WhatsApp                  Voice call +1 (919) 579-0479

10        10          Josh Hickman             vnd.android.cursor.item/com.google.android.apps.tachyo  (919) 579-0479
    1
                                               n.phone


10        10          Josh Hickman             vnd.android.cursor.item/com.google.android.apps.tachyo  (919) 579-0479

                                               n.phone.audio


┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ >....
.mode column

SELECT
    _id,
    datetime(date/1000, 'unixepoch') AS date_utc,
    duration,
    type,
    number
FROM calls
WHERE replace(replace(replace(replace(number, '+', ''), '-', ''), '(', ''), ')', '')
      LIKE '%9195790479'
ORDER BY date;
SQL

echo
echo "===== SMS MATCHES ====="
sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/contact_sms_correlation.txt
.headers on
.mode column

SELECT
    _id,
    datetime(date/1000, 'unixepoch') AS date_utc,
    type,
    address,
    length(body) AS body_length
FROM sms
WHERE replace(replace(replace(replace(address, '+', ''), '-', ''), '(', ''), ')', '')
      LIKE '%9195790479'
```

```
echo "═══ SMS MATCHES ═══"
sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/contact_sms_correlation.txt
.headers on
.mode column

SELECT
    _id,
    datetime(date/1000, 'unixepoch') AS date_utc,
    type,
    address,
    length(body) AS body_length
FROM sms
WHERE replace(replace(replace(replace(address, '+', ''), '-', ''), '(', ''), ')', '')
      LIKE '%9195790479'
ORDER BY date;
SQL
═══ CALL LOG MATCHES ═══
_id  date_utc              duration  type  number
___  ____________________  ________  ____  ____________
11   2020-02-14 01:29:01   11        2     9195790479
12   2020-02-14 01:30:24   65        1     +19195790479
13   2020-02-14 13:05:57   86        2     9195790479
14   2020-02-14 14:41:18   91        1     9195790479

═══ SMS MATCHES ═══
_id  date_utc              type  address       body_length
___  ____________________  ____  ____________  ___________
21   2020-02-09 18:41:42   2     9195790479    56
25   2020-02-14 13:09:03   2     919-579-0479  46

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/contact_mms_correlation.txt
.headers on
.mode column
.width 6 20 8 8 24 8

SELECT
    p._id AS mms_id,
    datetime(p.date, 'unixepoch') AS date_utc,
    p.msg_box,
    a.type AS address_type,
    a.address,
    p.m_type
FROM pdu AS p
LEFT JOIN addr AS a
    ON a.msg_id = p._id
WHERE replace(
        replace(
          replace(
            replace(a.address, '+', ''),
          '-', ''),
        '(', ''),
      ')', '')
      LIKE '%9195790479'
ORDER BY p.date, p._id, a.type;
SQL
mms_id  date_utc              msg_box  address_  address                  m_type
______  ___________________  _______  ________  _______________________  ______

1       2020-02-01 20:04:53  2        151       9195790479               128
```

```
    a.type AS address_type,
    a.address,
    p.m_type
FROM pdu AS p
LEFT JOIN addr AS a
    ON a.msg_id = p._id
WHERE replace(
        replace(
          replace(
            replace(a.address, '+', ''),
          '-', ''),
        '(', ''),
      ')', '')
      LIKE '%9195790479'
ORDER BY p.date, p._id, a.type;
SQL
```

mms_id	date_utc	msg_box	address_	address	m_type
1	2020-02-01 20:04:53	2	151	9195790479	128
2	2020-02-01 20:06:04	1	137	+19195790479	132
2	2020-02-01 20:06:04	1	151	+19195790479	132
3	2020-02-01 20:07:52	2	151	9195790479	128
4	2020-02-01 20:08:12	1	137	+19195790479	132
4	2020-02-01 20:08:12	1	151	+19195790479	132
5	2020-02-01 20:09:27	2	151	9195790479	128
6	2020-02-09 18:43:08	1	137	+19195790479	132
6	2020-02-09 18:43:08	1	151	+19195790479	132
7	2020-02-14 13:10:08	1	137	+19195790479	132
7	2020-02-14 13:10:08	1	151	+19195790479	132

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/contact_mms_text_correlation.txt
.headers on
.mode column

SELECT
    p._id AS mms_id,
    datetime(p.date, 'unixepoch') AS date_utc,
    p.msg_box,
    pt.ct AS content_type,
    length(pt.text) AS text_length,
    pt.text
FROM pdu AS p
JOIN part AS pt
    ON pt.mid = p._id
WHERE pt.ct = 'text/plain'
ORDER BY p.date, p._id;
SQL
mms_id  date_utc              msg_box  content_type  text_length  text
------  -------------------  -------  ------------  -----------  ----------------------------------------------------------------
1       2020-02-01 20:04:53  2        text/plain    15           Here's Signal.

2       2020-02-01 20:06:04  1        text/plain    70           It looks like the messaging is run through the Messages app.
                                                                   Strange.

3       2020-02-01 20:07:52  2        text/plain    47           It might be an attempt at unified messaging ... ?

4       2020-02-01 20:08:12  1        text/plain    6            Maybe?

5       2020-02-01 20:09:27  2        text/plain    61           It will be interesting to see where the messages are stored.
```

```
5       2020-02-01 20:09:27  2        text/plain    61          It will be interesting to see where the messages are stored.

6       2020-02-09 18:43:08  1        text/plain    13          No, thank you

7       2020-02-14 13:10:08  1        text/plain    20          That is interesting.

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ >....
        'SMS',
        _id,
        address,
        type,
        NULL,
        length(body)
    FROM sms

    UNION ALL

    SELECT
        datetime(p.date, 'unixepoch'),
        'MMS',
        p._id,
        (
            SELECT group_concat(DISTINCT a.address)
            FROM addr AS a
            WHERE a.msg_id = p._id
        ),
        p.msg_box,
        NULL,
        (
            SELECT SUM(length(pt.text))
            FROM part AS pt
            WHERE pt.mid = p._id
              AND pt.ct = 'text/plain'
        )
    FROM pdu AS p
)
```

```
    FROM pdu AS p
)
ORDER BY event_utc, artifact_type, record_id;
SQL
event_utc,artifact_type,record_id,participant,type_code,duration_seconds,content_length
"2020-01-29 18:20:45",SMS,2,+13233205525,1,,70
"2020-01-29 18:30:46",SMS,3,732873,1,,87
"2020-01-29 18:31:36",SMS,4,+18437514985,1,,55
"2020-01-29 18:32:08",SMS,5,+12253690855,1,,37
"2020-01-29 18:33:40",SMS,6,+12347570651,1,,37
"2020-01-29 18:35:03",SMS,7,+12349008124,1,,37
"2020-01-29 18:37:08",SMS,8,47543,1,,153
"2020-01-29 18:41:38",SMS,9,732873,1,,43
"2020-01-29 18:43:08",SMS,10,59109,1,,54
"2020-01-29 18:50:13",SMS,11,837402,1,,47
"2020-01-29 18:50:24",SMS,12,32665,1,,194
"2020-01-29 20:16:10",SMS,13,732873,1,,55
"2020-01-29 21:15:33",SMS,14,44398,1,,66
"2020-01-31 15:16:36",CALL,1,"",3,0,
"2020-01-31 18:07:50",CALL,2,"",3,0,
"2020-01-31 20:21:55",CALL,3,"",3,0,
"2020-02-01 01:00:03",SMS,15,+17162394257,1,,42
"2020-02-01 01:04:12",SMS,16,+13233208934,1,,88
"2020-02-01 19:28:00",CALL,4,"",3,0,
"2020-02-01 20:04:53",MMS,1,"+19195794674,9195790479",2,,15
"2020-02-01 20:06:04",MMS,2,+19195790479,1,,70
"2020-02-01 20:07:52",MMS,3,"+19195794674,9195790479",2,,47
"2020-02-01 20:08:12",MMS,4,+19195790479,1,,6
"2020-02-01 20:09:27",MMS,5,"+19195794674,9195790479",2,,61
"2020-02-03 19:18:46",SMS,17,32665,1,,86
"2020-02-03 19:32:22",CALL,5,"",3,0,
"2020-02-05 16:22:21",SMS,18,32665,1,,82
```

			29 18:32 :08			9K+9OFePKE9
6	8	+12347570651	2020-01- 29 18:33 :40	1	0	Your Skout verification code is 2661.
7	9	+12349008124	2020-01- 29 18:35 :03	1	0	Your Skout verification code is 9073.
8	10	47543	2020-01- 29 18:37 :08	1	0	<#> Your WhatsApp code: 362-593 You can also tap on this link to verify your phone: v.whatsapp.com/362 593 Don't share this code with others 4sgLq1p5sV6
9	5	732873	2020-01- 29 18:41 :38	1	0	Use 3649 as Microsoft account security code
10	11	59109	2020-01- 29 18:43 :08	1	0	[#][TikTok] 6327 is your verification code fJpzQvK2eu1
11	12	837402	2020-01- 29 18:50 :13	1	0	FB-55286 is ThisIs's Facebook confirmation code
12	13	32665	2020-01- 29 18:50 :24	1	0	Confirmed! To edit SMS preferences go to m.facebook.com/settings. To t urn off SMS for your Facebook account on this mobile number reply stop . Reply help for other options. SMS charges may apply.
13	5	732873	2020-01- 29 20:16 :10	1	0	<#> Your Imgur verification code is 4907230 WXYiuevlJ2b

13	5	732873	2020-01-29 20:16:10	1	0	<#> Your Imgur verification code is 4907230 WXYiuevlJ2b
14	14	44398	2020-01-29 21:15:33	1	0	369967 is your verification code for Dust - a safer place to text.
15	15	+17162394257	2020-02-01 01:00:03	1	1	Your Wickr Me confirmation code is 8434776
16	16	+13233208934	2020-02-01 01:04:12	1	1	Your Wire code is 172488. Open https://account.wire.com/v/172488 to verify your number.
17	13	32665	2020-02-03 19:18:46	1	0	👤 Xavier Mccollum is a new Facebook friend suggestion: https://fb.com/l/9OSKBTdMAcfo8C
18	13	32665	2020-02-05 16:22:21	1	0	🔔 ThisIs, you have 1 new notification on Facebook: https://fb.com/l/a5ANAeaZdoseLW
19	13	32665	2020-02-07 08:15:59	1	0	🔔 ThisIs, you have 1 new notification on Facebook: https://fb.com/l/1KFllApNpuN8igJ
20	13	32665	2020-02-08 18:17:01	1	0	🔔 ThisIs, you have 1 new notification on Facebook: https://fb.com/l/aym2DdgkRIfkix
21	17	9195790479	2020-02-09 18:41:42	2	1	I am on my way to the coffee shop. Do you want anything?
22	13	32665	2020-02-11 18:02	1	0	🔔 ThisIs, you have 6 new notifications on Facebook: https://fb.com/l/1IOHQ5bDdc60Nl9

```
17    13        32665                2020-02-03 19:18:46   1    0    👥 Xavier Mccollum is a new Facebook friend suggestion: https://fb.com/l/9OSKBTdMAcfo8C
18    13        32665                2020-02-05 16:22:21   1    0    🔔 ThisIs, you have 1 new notification on Facebook: https://fb.com/l/a5ANAeaZdoseLW
19    13        32665                2020-02-07 08:15:59   1    0    🔔 ThisIs, you have 1 new notification on Facebook: https://fb.com/l/1KFllApNpuN8igJ
20    13        32665                2020-02-08 18:17:01   1    0    🔔 ThisIs, you have 1 new notification on Facebook: https://fb.com/l/aym2DdgkRIfkix
21    17        9195790479           2020-02-09 18:41:42   2    1    I am on my way to the coffee shop. Do you want anything?
22    13        32665                2020-02-11 18:02:40   1    0    🔔 ThisIs, you have 6 new notifications on Facebook: https://fb.com/l/1IOHQ5bDdc60Nl9
23    13        32665                2020-02-12 19:12:18   1    0    🔔 ThisIs, you have 9 new notifications on Facebook: https://fb.com/l/akGyU7KWPrdRmS
24    13        32665                2020-02-13 21:12:24   1    0    🔔 ThisIs, you have 9 new notifications on Facebook: https://fb.com/l/1KypwvpROvYfv7n
25    17        919-579-0479         2020-02-14 13:09:03   2    1    Wow, the messages sync over between accounts.

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/mms_summary.txt
.headers on
.mode column
SELECT
    _id,
    thread_id,
    datetime(date, 'unixepoch') AS date_utc,
    msg_box,
    m_type,
    sub,
    m_size,
    ct_t
FROM pdu
ORDER BY date;
SQL
_id  thread_id  date_utc             msg_box  m_type  sub  m_size  ct_t
---  ---------  -------------------  -------  ------  ---  ------  ------------------------------------
1    17         2020-02-01 20:04:53  2        128             15      application/vnd.wap.multipart.related
2    17         2020-02-01 20:06:04  1        132
3    17         2020-02-01 20:07:52  2        128             47      application/vnd.wap.multipart.related
4    17         2020-02-01 20:08:12  1        132
5    17         2020-02-01 20:09:27  2        128             61      application/vnd.wap.multipart.related
6    17         2020-02-09 18:43:08  1        132
7    17         2020-02-14 13:10:08  1        132

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/mms_parts_and_addresses.txt
.headers on
.mode column
.width 5 20 8 8 22 6 7 32 20 20
SELECT
    p._id AS mms_id,
    datetime(p.date, 'unixepoch') AS date_utc,
    p.msg_box,
    p.m_type,
    a.address,
    a.type AS address_type,
    part._id AS part_id,
    part.ct AS content_type,
    part.name,
    part.fn
FROM pdu AS p
LEFT JOIN addr AS a ON a.msg_id = p._id
LEFT JOIN part ON part.mid = p._id
ORDER BY p.date, p._id, part._id;
SQL
```

mms_i	date_utc	msg_box	m_type	address	addres	part_id	content_type	name	fn
1	2020-02-01 20:04:53	2	128	+19195794674	137	1	application/smil		
1	2020-02-01 20:04:53	2	128	9195790479	151	1	application/smil		
1	2020-02-01 20:04:53	2	128	+19195794674	137	2	text/plain		
1	2020-02-01 20:04:53	2	128	9195790479	151	2	text/plain		
2	2020-02-01 20:06:04	1	132	+19195790479	137	3	text/plain	body	
2	2020-02-01 20:06:04	1	132	+19195790479	151	3	text/plain	body	

Session Actions Edit View Help

1	2020-02-01 20:04:53	2	128	+19195794674	137	2	text/plain	
1	2020-02-01 20:04:53	2	128	9195790479	151	2	text/plain	
2	2020-02-01 20:06:04	1	132	+19195790479	137	3	text/plain	body
2	2020-02-01 20:06:04	1	132	+19195790479	151	3	text/plain	body
3	2020-02-01 20:07:52	2	128	+19195794674	137	4	application/smil	
3	2020-02-01 20:07:52	2	128	9195790479	151	4	application/smil	
3	2020-02-01 20:07:52	2	128	+19195794674	137	5	text/plain	
3	2020-02-01 20:07:52	2	128	9195790479	151	5	text/plain	
4	2020-02-01 20:08:12	1	132	+19195790479	137	6	text/plain	body
4	2020-02-01 20:08:12	1	132	+19195790479	151	6	text/plain	body
5	2020-02-01 20:09:27	2	128	+19195794674	137	7	application/smil	
5	2020-02-01 20:09:27	2	128	9195790479	151	7	application/smil	
5	2020-02-01 20:09:27	2	128	+19195794674	137	8	text/plain	
5	2020-02-01 20:09:27	2	128	9195790479	151	8	text/plain	
6	2020-02-09 18:43:08	1	132	+19195790479	137	9	text/plain	body
6	2020-02-09 18:43:08	1	132	+19195790479	151	9	text/plain	body
7	2020-02-14 13:10:08	1	132	+19195790479	137	10	text/plain	body
7	2020-02-14 13:10:08	1	132	+19195790479	151	10	text/plain	body

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/mms_text_parts.txt
.headers on
.mode column
SELECT
    p._id AS mms_id,
    datetime(p.date, 'unixepoch') AS date_utc,
    p.msg_box,
    part._id AS part_id,
    part.ct AS content_type,
    part._data,
    part.text
FROM pdu AS p
JOIN part ON part.mid = p._id
WHERE part.ct = 'text/plain'
ORDER BY p.date, p._id, part._id;
SQL
mms_id  date_utc              msg_box  part_id  content_type  _data  text
------  -------------------  -------  -------  ------------  -----  ----------------------------------------------------------
1       2020-02-01 20:04:53  2        2        text/plain           Here's Signal.

2       2020-02-01 20:06:04  1        3        text/plain           It looks like the messaging is run through the Messages app.
                                                                    Strange.

3       2020-02-01 20:07:52  2        5        text/plain           It might be an attempt at unified messaging ... ?

4       2020-02-01 20:08:12  1        6        text/plain           Maybe?

5       2020-02-01 20:09:27  2        8        text/plain           It will be interesting to see where the messages are stored.


6       2020-02-09 18:43:08  1        9        text/plain           No, thank you

7       2020-02-14 13:10:08  1        10       text/plain           That is interesting.

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

find "working/extracted/Pixel 3/data" \
  -type f \( \
    -name "contacts2.db" -o \
    -name "calllog.db" -o \
    -name "calendar.db" \
  \) -print | tee outputs/contact_call_calendar_paths.txt
working/extracted/Pixel 3/data/data/com.android.providers.calendar/databases/calendar.db
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

mkdir -p working/databases

cp -p \
  "working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db" \
  working/databases/contacts2.db

cp -p \
  "working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db" \
  working/databases/calllog.db

cp -p \
  "working/extracted/Pixel 3/data/data/com.android.providers.calendar/databases/calendar.db" \
  working/databases/calendar.db

sha256sum \
  "working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db" \
  working/databases/contacts2.db \
  "working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db" \
  working/databases/calllog.db \
  "working/extracted/Pixel 3/data/data/com.android.providers.calendar/databases/calendar.db" \
  working/databases/calendar.db \
  | tee outputs/contact_call_calendar_hashes.txt
2a1fee3d9412e2f746592fcc0f7abcdd0ece26bc02c95240fd210464f4625050  working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db
2a1fee3d9412e2f746592fcc0f7abcdd0ece26bc02c95240fd210464f4625050  working/databases/contacts2.db
5979a3fa0b4e70aff6dc20675e84a14fa584e52ad4d224d9da14d8aa0ef809d3  working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db
5979a3fa0b4e70aff6dc20675e84a14fa584e52ad4d224d9da14d8aa0ef809d3  working/databases/calllog.db
82a580901fca7580dd8841a1c988dcaa65afef068f386d603aeec6039e5aca3c  working/extracted/Pixel 3/data/data/com.android.providers.calendar/databases/calendar.db
82a580901fca7580dd8841a1c988dcaa65afef068f386d603aeec6039e5aca3c  working/databases/calendar.db
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

for db in contacts2.db calllog.db calendar.db; do
  echo "═══ $db ═══"
  sqlite3 "working/databases/$db" \
    "PRAGMA integrity_check; .tables"
done 2>&1 | tee outputs/contact_call_calendar_schema.txt
═══ contacts2.db ═══
Error: in prepare, no such collation sequence: PHONEBOOK
═══ calllog.db ═══
Error: in prepare, near ".": syntax error
  .tables
  ^--- error here
ok
═══ calendar.db ═══
Error: in prepare, near ".": syntax error
  .tables
  ^--- error here
ok
```

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

for db in contacts2.db calllog.db calendar.db; do
  echo "===== $db ====="
  sqlite3 "working/databases/$db" <<'SQL'
PRAGMA integrity_check;
.tables
SQL
done | tee outputs/contact_call_calendar_schema.txt
===== contacts2.db =====
Parse error near line 1: no such collation sequence: PHONEBOOK
_sync_state               phone_lookup              view_data_usage
_sync_state_metadata      photo_files               view_entities
accounts                  pre_authorized_uris       view_groups
agg_exceptions            presence                  view_metadata_sync
agg_presence              properties                view_metadata_sync_state
android_metadata          raw_contacts              view_raw_contacts
contacts                  search_index              view_raw_entities
data                      search_index_content      view_stream_items
data_usage_stat           search_index_docsize      view_v1_contact_methods
default_directory         search_index_segdir       view_v1_extensions
deleted_contacts          search_index_segments     view_v1_group_membership
directories               search_index_stat         view_v1_groups
groups                    settings                  view_v1_organizations
metadata_sync             status_updates            view_v1_people
metadata_sync_state       stream_item_photos        view_v1_phones
mimetypes                 stream_items              view_v1_photos
name_lookup               v1_settings               visible_contacts
nickname_lookup           view_contacts
packages                  view_data
===== calllog.db =====
ok
android_metadata  calls             properties        voicemail_status
===== calendar.db =====
ok
Attendees             Colors                Reminders
CalendarAlerts        Events                _sync_state
CalendarCache         EventsRawTimes        _sync_state_metadata
```

```
==== calllog.db ====
ok
android_metadata  calls             properties        voicemail_status
==== calendar.db ====
ok
Attendees             Colors                Reminders
CalendarAlerts        Events                _sync_state
CalendarCache         EventsRawTimes        _sync_state_metadata
CalendarMetaData      ExtendedProperties    android_metadata
Calendars             Instances             view_events

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/calllog.db <<'SQL' | tee outputs/calllog_schema.txt
PRAGMA integrity_check;
.schema calls
SQL
ok
CREATE TABLE calls (_id INTEGER PRIMARY KEY AUTOINCREMENT,number TEXT,presentation INTEGER NOT NULL DEFAULT 1,post_dial_digits TEXT NOT NULL DEFAULT '',via_number
 TEXT NOT NULL DEFAULT '',date INTEGER,duration INTEGER,data_usage INTEGER,type INTEGER,features INTEGER NOT NULL DEFAULT 0,subscription_component_name TEXT,subsc
ription_id TEXT,phone_account_address TEXT,phone_account_hidden INTEGER NOT NULL DEFAULT 0,sub_id INTEGER DEFAULT -1,new INTEGER,name TEXT,numbertype INTEGER,numb
erlabel TEXT,countryiso TEXT,voicemail_uri TEXT,is_read INTEGER,geocoded_location TEXT,lookup_uri TEXT,matched_number TEXT,normalized_number TEXT,photo_id INTEGER
 NOT NULL DEFAULT 0,photo_uri TEXT,formatted_number TEXT,add_for_all_users INTEGER NOT NULL DEFAULT 1,last_modified INTEGER DEFAULT 0,call_screening_component_nam
e TEXT,call_screening_app_name TEXT,block_reason INTEGER NOT NULL DEFAULT 0,_data TEXT,has_content INTEGER,mime_type TEXT,source_data TEXT,source_package TEXT,tra
nscription TEXT,transcription_state INTEGER NOT NULL DEFAULT 0,state INTEGER,dirty INTEGER NOT NULL DEFAULT 0,deleted INTEGER NOT NULL DEFAULT 0,backed_up INTEGER
 NOT NULL DEFAULT 0,restored INTEGER NOT NULL DEFAULT 0,archived INTEGER NOT NULL DEFAULT 0,is_omtp_voicemail INTEGER NOT NULL DEFAULT 0);

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/calllog.db <<'SQL' | tee outputs/calllog_summary.txt
.headers on
.mode column
SELECT
    _id,
    date,
    datetime(date/1000, 'unixepoch') AS date_utc,
    duration,
    type,
    new,
    number,
    name,
    geocoded_location
FROM calls
ORDER BY date;
SQL
_id  date           date_utc             duration  type  new  number         name          geocoded_location
---  -------------  -------------------  --------  ----  ---  -------------  ------------  -----------------
1    1580483796007  2020-01-31 15:16:36  0         3     0
2    1580494070836  2020-01-31 18:07:50  0         3     0
```

```
nscription TEXT,transcription_state INTEGER NOT NULL DEFAULT 0,state INTEGER,dirty INTEGER NOT NULL DEFAULT 0,deleted INTEGER NOT NULL DEFAULT 0,backed_up INTEGER
 NOT NULL DEFAULT 0,restored INTEGER NOT NULL DEFAULT 0,archived INTEGER NOT NULL DEFAULT 0,is_omtp_voicemail INTEGER NOT NULL DEFAULT 0);

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/calllog.db <<'SQL' | tee outputs/calllog_summary.txt
.headers on
.mode column
SELECT
    _id,
    date,
    datetime(date/1000, 'unixepoch') AS date_utc,
    duration,
    type,
    new,
    number,
    name,
    geocoded_location
FROM calls
ORDER BY date;
SQL
_id  date           date_utc             duration  type  new  number          name          geocoded_location
___  _____________  ___________________  ________  ____  ___  ______________  ____________  _________________
1    1580483796007  2020-01-31 15:16:36  0         3     0
2    1580494070836  2020-01-31 18:07:50  0         3     0
3    1580502115252  2020-01-31 20:21:55  0         3     0
4    1580585280414  2020-02-01 19:28:00  0         3     0
5    1580758342425  2020-02-03 19:32:22  0         3     0
6    1581192896305  2020-02-08 20:14:56  0         3     0    +332629243087
7    1581521849939  2020-02-12 15:37:29  0         3     0    +19195121037                North Carolina
8    1581521901000  2020-02-12 15:38:21  3         4     0    +19195121037                North Carolina
9    1581555164000  2020-02-13 00:52:44  0         3     0    +15755863247                Questa, NM
10   1581555213000  2020-02-13 00:53:33  3         4     0    +15755863247                Questa, NM
11   1581643741051  2020-02-14 01:29:01  11        2     1    9195790479      Josh Hickman  North Carolina
12   1581643824061  2020-02-14 01:30:24  65        1     1    +19195790479    Josh Hickman  North Carolina
13   1581685557104  2020-02-14 13:05:57  86        2     1    9195790479      Josh Hickman  North Carolina
14   1581691278351  2020-02-14 14:41:18  91        1     1    9195790479      Josh Hickman  North Carolina

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
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```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/calendar.db <<'SQL' | tee outputs/calendar_schema.txt
PRAGMA integrity_check;
.schema Events
.schema Calendars
.schema Attendees
SQL
ok
CREATE TABLE Events (_id INTEGER PRIMARY KEY AUTOINCREMENT,_sync_id TEXT,dirty INTEGER,mutators TEXT,lastSynced INTEGER DEFAULT 0,calendar_id INTEGER NOT NULL,title TEXT,eventLocation TEXT,description TEXT,eventColor INTEGER,eventColor_index TEXT,eventStatus INTEGER,selfAttendeeStatus INTEGER NOT NULL DEFAULT 0,dtstart INTEGER,dtend INTEGER,eventTimezone TEXT,duration TEXT,allDay INTEGER NOT NULL DEFAULT 0,accessLevel INTEGER NOT NULL DEFAULT 0,availability INTEGER NOT NULL DEFAULT 0,hasAlarm INTEGER NOT NULL DEFAULT 0,hasExtendedProperties INTEGER NOT NULL DEFAULT 0,rrule TEXT,rdate TEXT,exrule TEXT,exdate TEXT,original_id INTEGER,original_sync_id TEXT,originalInstanceTime INTEGER,originalAllDay INTEGER,lastDate INTEGER,hasAttendeeData INTEGER NOT NULL DEFAULT 0,guestsCanModify INTEGER NOT NULL DEFAULT 0,guestsCanInviteOthers INTEGER NOT NULL DEFAULT 1,guestsCanSeeGuests INTEGER NOT NULL DEFAULT 1,organizer STRING,isOrganizer INTEGER,deleted INTEGER NOT NULL DEFAULT 0,eventEndTimezone TEXT,customAppPackage TEXT,customAppUri TEXT,uid2445 TEXT,sync_data1 TEXT,sync_data2 TEXT,sync_data3 TEXT,sync_data4 TEXT,sync_data5 TEXT,sync_data6 TEXT,sync_data7 TEXT,sync_data8 TEXT,sync_data9 TEXT,sync_data10 TEXT);
CREATE INDEX eventsCalendarIdIndex ON Events (calendar_id);
CREATE TRIGGER events_cleanup_delete DELETE ON Events BEGIN DELETE FROM Instances WHERE event_id=old._id;DELETE FROM EventsRawTimes WHERE event_id=old._id;DELETE FROM Attendees WHERE event_id=old._id;DELETE FROM Reminders WHERE event_id=old._id;DELETE FROM CalendarAlerts WHERE event_id=old._id;DELETE FROM ExtendedProperties WHERE event_id=old._id;END;
CREATE TRIGGER event_color_update UPDATE OF eventColor_index ON Events WHEN new.eventColor_index NOT NULL BEGIN UPDATE Events SET eventColor=(SELECT color FROM Colors WHERE account_name=(SELECT account_name FROM Calendars WHERE _id=new.calendar_id) AND account_type=(SELECT account_type FROM Calendars WHERE _id=new.calendar_id) AND color_index=new.eventColor_index AND color_type=1)  WHERE _id=old._id; END;
CREATE TRIGGER original_sync_update UPDATE OF _sync_id ON Events BEGIN UPDATE Events SET original_sync_id=new._sync_id WHERE original_id=old._id; END;
CREATE TABLE Calendars (_id INTEGER PRIMARY KEY,account_name TEXT,account_type TEXT,_sync_id TEXT,dirty INTEGER,mutators TEXT,name TEXT,calendar_displayName TEXT,calendar_color INTEGER,calendar_color_index TEXT,calendar_access_level INTEGER,visible INTEGER NOT NULL DEFAULT 1,sync_events INTEGER NOT NULL DEFAULT 0,calendar_location TEXT,calendar_timezone TEXT,ownerAccount TEXT, isPrimary INTEGER, canOrganizerRespond INTEGER NOT NULL DEFAULT 1,canModifyTimeZone INTEGER DEFAULT 1,canPartiallyUpdate INTEGER DEFAULT 0,maxReminders INTEGER DEFAULT 5,allowedReminders TEXT DEFAULT '0,1',allowedAvailability TEXT DEFAULT '0,1',allowedAttendeeTypes TEXT DEFAULT '0,1,2',deleted INTEGER NOT NULL DEFAULT 0,cal_sync1 TEXT,cal_sync2 TEXT,cal_sync3 TEXT,cal_sync4 TEXT,cal_sync5 TEXT,cal_sync6 TEXT,cal_sync7 TEXT,cal_sync8 TEXT,cal_sync9 TEXT,cal_sync10 TEXT);
CREATE TRIGGER calendar_cleanup DELETE ON Calendars BEGIN DELETE FROM Events WHERE calendar_id=old._id;END;
CREATE TRIGGER calendar_color_update UPDATE OF calendar_color_index ON Calendars WHEN new.calendar_color_index NOT NULL BEGIN UPDATE Calendars SET calendar_color=(SELECT color FROM Colors WHERE account_name=new.account_name AND account_type=new.account_type AND color_index=new.calendar_color_index AND color_type=0)  WHERE _id=old._id; END;
CREATE TABLE Attendees (_id INTEGER PRIMARY KEY,event_id INTEGER,attendeeName TEXT,attendeeEmail TEXT,attendeeStatus INTEGER,attendeeRelationship INTEGER,attendeeType INTEGER,attendeeIdentity TEXT,attendeeIdNamespace TEXT);
```

```
CREATE TRIGGER original_sync_update UPDATE OF _sync_id ON Events BEGIN UPDATE Events SET original_sync_id=new._sync_id WHERE original_id=old._id; END;
CREATE TABLE Calendars (_id INTEGER PRIMARY KEY,account_name TEXT,account_type TEXT,_sync_id TEXT,dirty INTEGER,mutators TEXT,name TEXT,calendar_displayName TEXT,
calendar_color INTEGER,calendar_color_index TEXT,calendar_access_level INTEGER,visible INTEGER NOT NULL DEFAULT 1,sync_events INTEGER NOT NULL DEFAULT 0,calendar_
location TEXT,calendar_timezone TEXT,ownerAccount TEXT, isPrimary INTEGER, canOrganizerRespond INTEGER NOT NULL DEFAULT 1,canModifyTimeZone INTEGER DEFAULT 1,canP
artiallyUpdate INTEGER DEFAULT 0,maxReminders INTEGER DEFAULT 5,allowedReminders TEXT DEFAULT '0,1',allowedAvailability TEXT DEFAULT '0,1',allowedAttendeeTypes TE
XT DEFAULT '0,1,2',deleted INTEGER NOT NULL DEFAULT 0,cal_sync1 TEXT,cal_sync2 TEXT,cal_sync3 TEXT,cal_sync4 TEXT,cal_sync5 TEXT,cal_sync6 TEXT,cal_sync7 TEXT,cal
_sync8 TEXT,cal_sync9 TEXT,cal_sync10 TEXT);
CREATE TRIGGER calendar_cleanup DELETE ON Calendars BEGIN DELETE FROM Events WHERE calendar_id=old._id;END;
CREATE TRIGGER calendar_color_update UPDATE OF calendar_color_index ON Calendars WHEN new.calendar_color_index NOT NULL BEGIN UPDATE Calendars SET calendar_color=
(SELECT color FROM Colors WHERE account_name=new.account_name AND account_type=new.account_type AND color_index=new.calendar_color_index AND color_type=0)  WHERE
_id=old._id; END;
CREATE TABLE Attendees (_id INTEGER PRIMARY KEY,event_id INTEGER,attendeeName TEXT,attendeeEmail TEXT,attendeeStatus INTEGER,attendeeRelationship INTEGER,attendee
Type INTEGER,attendeeIdentity TEXT,attendeeIdNamespace TEXT);
CREATE INDEX attendeesEventIdIndex ON Attendees (event_id);

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/calendar.db <<'SQL' | tee outputs/calendar_events.txt
.headers on
.mode column

SELECT *
FROM Events
LIMIT 10;
SQL

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/calendar.db <<'SQL' | tee outputs/calendar_counts.txt
.headers on
.mode column

SELECT 'Events' AS table_name, COUNT(*) AS record_count FROM Events;
SELECT 'Calendars' AS table_name, COUNT(*) AS record_count FROM Calendars;
SELECT 'Attendees' AS table_name, COUNT(*) AS record_count FROM Attendees;
SELECT 'Reminders' AS table_name, COUNT(*) AS record_count FROM Reminders;
SQL
table_name  record_count
──────────  ────────────
Events      0
table_name  record_count
──────────  ────────────
Calendars   1
table_name  record_count
──────────  ────────────
Attendees   0
table_name  record_count
──────────  ────────────
Reminders   0

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/contacts2.db <<'SQL' 2>&1 | tee outputs/contacts_inspection.txt
.headers on
.mode column

SELECT name, type
FROM sqlite_master
WHERE type IN ('table','view')
  AND name IN ('contacts','raw_contacts','data','mimetypes');

SELECT COUNT(*) AS contacts_count FROM contacts;
SELECT COUNT(*) AS raw_contacts_count FROM raw_contacts;
SELECT COUNT(*) AS data_count FROM data;
SELECT COUNT(*) AS mimetypes_count FROM mimetypes;
SQL
name          type
------------  -----
contacts      table
raw_contacts  table
mimetypes     table
data          table
contacts_count
--------------
1
raw_contacts_count
------------------
10
data_count
----------
35
mimetypes_count
---------------
30

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/calendar.db <<'SQL' | tee outputs/calendar_account_details.txt
.headers on
.mode column

SELECT
    _id,
    account_name,
    account_type,
    name,
    calendar_displayName,
    visible,
    sync_events,
    ownerAccount,
    isPrimary,
    deleted
FROM Calendars;
SQL
_id  account_name          account_type  name                calendar_displayName  visible  sync_events  ownerAccount          isPrimary  deleted
---  -------------------   ------------  ------------------  --------------------  -------  -----------  --------------------  ---------  -------
1    thisisdfir@gmail.com  com.google    thisisdfir@gmail.com  thisisdfir@gmail.com  1        1            thisisdfir@gmail.com  1          0

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/mmssms_schema.txt
PRAGMA integrity_check;
.tables
.schema sms
.schema pdu
.schema part
.schema addr
SQL
ok
addr                pdu_restricted      threads
android_metadata    pending_msgs        words
attachments         rate                words_content
canonical_addresses raw                 words_segdir
drm                 sms                 words_segments
part                sms_restricted
pdu                 sr_pending
CREATE TABLE sms (_id INTEGER PRIMARY KEY,thread_id INTEGER,address TEXT,person INTEGER,date INTEGER,date_sent INTEGER DEFAULT 0,protocol INT
EGER,read INTEGER DEFAULT 0,status INTEGER DEFAULT -1,type INTEGER,reply_path_present INTEGER,subject TEXT,body TEXT,service_center TEXT,lock
ed INTEGER DEFAULT 0,sub_id INTEGER DEFAULT -1, error_code INTEGER DEFAULT 0,creator TEXT,seen INTEGER DEFAULT 0);
CREATE TRIGGER sms_update_thread_on_insert AFTER INSERT ON sms BEGIN  UPDATE threads SET    date = (strftime('%s','now') * 1000),     snippet
 = new.body,     snippet_cs = 0  WHERE threads._id = new.thread_id;   UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms
  LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id        AND sms.type != 3) +      (SELECT COUNT(pdu._
id) FROM pdu LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id        AND (m_type=132 OR m_type=130 OR
m_type=128)        AND msg_box != 3)   WHERE threads._id = new.thread_id;   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM
 sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id
; END;
CREATE TRIGGER sms_update_thread_date_subject_on_update AFTER  UPDATE OF date, body, type  ON sms BEGIN  UPDATE threads SET    date = (strfti
me('%s','now') * 1000),     snippet = new.body,     snippet_cs = 0  WHERE threads._id = new.thread_id;   UPDATE threads SET message_count =
    (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id        AND sms.ty
pe != 3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id
  AND (m_type=132 OR m_type=130 OR m_type=128)        AND msg_box != 3)   WHERE threads._id = new.thread_id;   UPDATE threads SET read =
CASE (SELECT COUNT(*)          FROM sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END
  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER sms_update_thread_read_on_update AFTER  UPDATE OF read  ON sms BEGIN   UPDATE threads SET read =     CASE (SELECT COUNT(*)
      FROM sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new
.thread_id; END;
CREATE TRIGGER update_threads_error_on_update_sms   AFTER UPDATE OF type ON sms  WHEN (OLD.type != 5 AND NEW.type = 5)    OR (OLD.type = 5 AN
D NEW.type != 5) BEGIN   UPDATE threads SET error =     CASE      WHEN NEW.type = 5 THEN error + 1      ELSE error - 1    END   WHERE _id = N
```

```
CASE (SELECT COUNT(*)          FROM sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END
  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER sms_update_thread_read_on_update AFTER  UPDATE OF read  ON sms BEGIN   UPDATE threads SET read =     CASE (SELECT COUNT(*)
      FROM sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new
.thread_id; END;
CREATE TRIGGER update_threads_error_on_update_sms   AFTER UPDATE OF type ON sms  WHEN (OLD.type =≠ 5 AND NEW.type = 5)    OR (OLD.type = 5 AN
D NEW.type =≠ 5) BEGIN   UPDATE threads SET error =     CASE      WHEN NEW.type = 5 THEN error + 1      ELSE error - 1    END   WHERE _id = N
EW.thread_id; END;
CREATE TRIGGER sms_words_update AFTER UPDATE ON sms BEGIN UPDATE words  SET index_text = NEW.body WHERE (source_id=NEW._id AND table_to_use=1
);  END;
CREATE TRIGGER sms_words_delete AFTER DELETE ON sms BEGIN DELETE FROM   words WHERE source_id = OLD._id AND table_to_use = 1; END;
CREATE INDEX typeThreadIdIndex ON sms (type, thread_id);
CREATE INDEX threadIdDateIndex ON sms (thread_id, date);
CREATE TABLE pdu (_id INTEGER PRIMARY KEY AUTOINCREMENT,thread_id INTEGER,date INTEGER,date_sent INTEGER DEFAULT 0,msg_box INTEGER,read INTEG
ER DEFAULT 0,m_id TEXT,sub TEXT,sub_cs INTEGER,ct_t TEXT,ct_l TEXT,exp INTEGER,m_cls TEXT,m_type INTEGER,v INTEGER,m_size INTEGER,pri INTEGER
,rr INTEGER,rpt_a INTEGER,resp_st INTEGER,st INTEGER,tr_id TEXT,retr_st INTEGER,retr_txt TEXT,retr_txt_cs INTEGER,read_status INTEGER,ct_cls
INTEGER,resp_txt TEXT,d_tm INTEGER,d_rpt INTEGER,locked INTEGER DEFAULT 0,sub_id INTEGER DEFAULT -1, seen INTEGER DEFAULT 0,creator TEXT,text
_only INTEGER DEFAULT 0);
CREATE TRIGGER part_cleanup DELETE ON pdu BEGIN   DELETE FROM part  WHERE mid=old._id;END;
CREATE TRIGGER addr_cleanup DELETE ON pdu BEGIN   DELETE FROM addr  WHERE msg_id=old._id;END;
CREATE TRIGGER cleanup_delivery_and_read_report AFTER DELETE ON pdu WHEN old.m_type=128 BEGIN   DELETE FROM pdu  WHERE (m_type=134    OR m_ty
pe=136)    AND m_id=old.m_id; END;
CREATE TRIGGER update_threads_on_update_pdu  AFTER UPDATE of thread_id ON pdu  BEGIN   UPDATE threads SET has_attachment=1 WHERE _id IN    (S
ELECT pdu.thread_id FROM part JOIN pdu      WHERE part.ct =≠ 'text/plain' AND part.ct =≠ 'application/smil'      AND part.mid = pdu._id); END
;
CREATE TRIGGER delete_mms_pending_on_delete AFTER DELETE ON pdu BEGIN   DELETE FROM pending_msgs  WHERE msg_id=old._id; END;
CREATE TRIGGER delete_mms_pending_on_update AFTER UPDATE ON pdu WHEN old.msg_box=4  AND new.msg_box=≠4 BEGIN   DELETE FROM pending_msgs  WHER
E msg_id=new._id; END;
CREATE TRIGGER insert_mms_pending_on_insert AFTER INSERT ON pdu WHEN new.m_type=130  OR new.m_type=135 BEGIN   INSERT INTO pending_msgs    (p
roto_type,     msg_id,     msg_type,     err_type,     err_code,     retry_index,     due_time)   VALUES     (1,      new._id,      new.m_typ
e,0,0,0,0);END;
CREATE TRIGGER insert_mms_pending_on_update AFTER UPDATE ON pdu WHEN new.m_type=128  AND new.msg_box=4  AND old.msg_box=≠4 BEGIN   INSERT INT
O pending_msgs    (proto_type,     msg_id,     msg_type,     err_type,     err_code,     retry_index,     due_time)   VALUES     (1,      new
._id,      new.m_type,0,0,0,0);END;
CREATE TRIGGER pdu_update_thread_date_subject_on_update AFTER  UPDATE OF date, sub, msg_box  ON pdu   WHEN new.m_type=132    OR new.m_type=13
0    OR new.m_type=128 BEGIN  UPDATE threads SET    date = (strftime('%s','now') * 1000),     snippet = new.sub,     snippet_cs = new.sub_cs
 WHERE threads._id = new.thread_id;   UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON thre
ads._id = thread_id      WHERE thread_id = new.thread_id        AND sms.type =≠ 3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads
     ON threads._id = thread_id      WHERE thread_id = new.thread_id        AND (m_type=132 OR m_type=130 OR m_type=128)        AND msg_box !
```

```
    ON threads._id = thread_id      WHERE thread_id = new.thread_id        AND (m_type=132 OR m_type=130 OR m_type=128)        AND msg_box !
= 3)   WHERE threads._id = new.thread_id;   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM pdu          WHERE read = 0
      AND thread_id = threads._id             AND (m_type=132 OR m_type=130 OR m_type=128))       WHEN 0 THEN 1      ELSE 0    END  WHERE t
hreads._id = new.thread_id; END;
CREATE TRIGGER pdu_update_thread_on_delete AFTER DELETE ON pdu BEGIN   UPDATE threads SET      date = (strftime('%s','now') * 1000)  WHERE th
reads._id = old.thread_id;   UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON threads._id =
 thread_id      WHERE thread_id = old.thread_id        AND sms.type ≠ 3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads       ON t
hreads._id = thread_id      WHERE thread_id = old.thread_id        AND (m_type=132 OR m_type=130 OR m_type=128)        AND msg_box ≠ 3)   WH
ERE threads._id = old.thread_id;   UPDATE threads SET snippet =    (SELECT snippet FROM     (SELECT date * 1000 AS date, sub AS snippet, thre
ad_id FROM pdu      UNION SELECT date, body AS snippet, thread_id FROM sms)    WHERE thread_id = OLD.thread_id ORDER BY date DESC LIMIT 1)
WHERE threads._id = OLD.thread_id;   UPDATE threads SET snippet_cs =    (SELECT snippet_cs FROM     (SELECT date * 1000 AS date, sub_cs AS sn
ippet_cs, thread_id FROM pdu      UNION SELECT date, 0 AS snippet_cs, thread_id FROM sms)    WHERE thread_id = OLD.thread_id ORDER BY date DE
SC LIMIT 1)   WHERE threads._id = OLD.thread_id; END;
CREATE TRIGGER pdu_update_thread_on_insert AFTER INSERT ON pdu   WHEN new.m_type=132    OR new.m_type=130    OR new.m_type=128 BEGIN  UPDATE
threads SET    date = (strftime('%s','now') * 1000),     snippet = new.sub,     snippet_cs = new.sub_cs  WHERE threads._id = new.thread_id;
 UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON threads._id = thread_id      WHERE thread
_id = new.thread_id        AND sms.type ≠ 3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads       ON threads._id = thread_id
WHERE thread_id = new.thread_id        AND (m_type=132 OR m_type=130 OR m_type=128)        AND msg_box ≠ 3)   WHERE threads._id = new.thread
_id;   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM pdu          WHERE read = 0          AND thread_id = threads._id
          AND (m_type=132 OR m_type=130 OR m_type=128))       WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER pdu_update_thread_read_on_update AFTER  UPDATE OF read  ON pdu   WHEN new.m_type=132    OR new.m_type=130    OR new.m_type=128
 BEGIN   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM pdu          WHERE read = 0          AND thread_id = threads._id
            AND (m_type=132 OR m_type=130 OR m_type=128))       WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER update_threads_error_on_delete_mms   BEFORE DELETE ON pdu  WHEN OLD._id IN (SELECT DISTINCT msg_id                  FROM pend
ing_msgs                  WHERE err_type ⩾ 10) BEGIN   UPDATE threads SET error = error - 1  WHERE _id = OLD.thread_id; END;
CREATE TRIGGER update_threads_error_on_move_mms   BEFORE UPDATE OF msg_box ON pdu   WHEN (OLD.msg_box = 4 AND NEW.msg_box ≠ 4)   AND (OLD._i
d IN (SELECT DISTINCT msg_id                  FROM pending_msgs                  WHERE err_type ⩾ 10)) BEGIN   UPDATE threads SET error =
error - 1  WHERE _id = OLD.thread_id; END;
CREATE TABLE part (_id INTEGER PRIMARY KEY AUTOINCREMENT,mid INTEGER,seq INTEGER DEFAULT 0,ct TEXT,name TEXT,chset INTEGER,cd TEXT,fn TEXT,ci
d TEXT,cl TEXT,ctt_s INTEGER,ctt_t TEXT,_data TEXT,text TEXT);
CREATE TRIGGER update_threads_on_insert_part  AFTER INSERT ON part  WHEN new.ct ≠ 'text/plain' AND new.ct ≠ 'application/smil'  BEGIN   UPD
ATE threads SET has_attachment=1 WHERE _id IN    (SELECT pdu.thread_id FROM part JOIN pdu ON pdu._id=part.mid      WHERE part._id=new._id LIM
IT 1);  END;
CREATE TRIGGER update_threads_on_update_part  AFTER UPDATE of mid ON part  WHEN new.ct ≠ 'text/plain' AND new.ct ≠ 'application/smil'  BEGI
N   UPDATE threads SET has_attachment=1 WHERE _id IN    (SELECT pdu.thread_id FROM part JOIN pdu ON pdu._id=part.mid      WHERE part._id=new.
_id LIMIT 1);  END;
CREATE TRIGGER update_threads_on_delete_part  AFTER DELETE ON part  WHEN old.ct ≠ 'text/plain' AND old.ct ≠ 'application/smil'  BEGIN   UPD
ATE threads SET has_attachment =    CASE     (SELECT COUNT(*) FROM part JOIN pdu      WHERE pdu.thread_id = threads._id      AND part.ct ≠ '
text/plain' AND part.ct ≠ 'application/smil'      AND part.mid = pdu._id)   WHEN 0 THEN 0    ELSE 1    END;  END;
```

```
CREATE TRIGGER update_threads_on_delete_part  AFTER DELETE ON part  WHEN old.ct ≠ 'text/plain' AND old.ct ≠ 'application/smil'  BEGIN   UPD
ATE threads SET has_attachment =    CASE     (SELECT COUNT(*) FROM part JOIN pdu      WHERE pdu.thread_id = threads._id      AND part.ct ≠ '
text/plain' AND part.ct ≠ 'application/smil'      AND part.mid = pdu._id)   WHEN 0 THEN 0    ELSE 1    END;  END;
CREATE TRIGGER mms_words_update AFTER UPDATE ON part BEGIN UPDATE words  SET index_text = NEW.text WHERE (source_id=NEW._id AND table_to_use=
2);  END;
CREATE TRIGGER mms_words_delete AFTER DELETE ON part BEGIN DELETE FROM  words WHERE source_id = OLD._id AND table_to_use = 2; END;
CREATE INDEX partMidIndex ON part (mid);
CREATE TABLE addr (_id INTEGER PRIMARY KEY,msg_id INTEGER,contact_id INTEGER,address TEXT,type INTEGER,charset INTEGER);
CREATE INDEX addrMsgIdIndex ON addr (msg_id);
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

find "working/extracted/Pixel 3/data" -type f \
  \( -iname "*.db" -o -iname "*.sqlite" -o -iname "*.sqlite3" \) \
  -printf '%p\t%s bytes\n' \
  | sort \
  | tee outputs/database_inventory.txt
working/extracted/Pixel 3/data/adb/magisk.db    40960 bytes
working/extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/databases/Databases.db        32768 bytes
working/extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/heavy_ad_intervention_opt_out.db      20480 bytes
working/extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/Offline Pages/metadata/OfflinePages.db        24576 bytes
working/extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/Offline Pages/request_queue/RequestQueue.db   12288 bytes
working/extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/page_load_capping_opt_out.db  20480 bytes
working/extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/previews_opt_out.db   20480 bytes
working/extracted/Pixel 3/data/data/com.android.chrome/app_chrome/Default/Sync Data/SyncData.sqlite3    393216 bytes
working/extracted/Pixel 3/data/data/com.android.keychain/databases/grants.db    28672 bytes
working/extracted/Pixel 3/data/data/com.android.providers.calendar/databases/calendar.db        122880 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db 40960 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db       376832 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/profile.db 376832 bytes
working/extracted/Pixel 3/data/data/com.android.providers.downloads/databases/downloads.db      24576 bytes
working/extracted/Pixel 3/data/data/com.android.providers.media/databases/external.db   212992 bytes
working/extracted/Pixel 3/data/data/com.android.providers.media/databases/internal.db   163840 bytes
working/extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db 118784 bytes
working/extracted/Pixel 3/data/data/com.android.providers.userdictionary/databases/user_dict.db 16384 bytes
working/extracted/Pixel 3/data/data/com.android.service.ims.presence/databases/rcseab.db        20480 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/auto_update.db        32768 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/counters.db   20480 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/data_usage.db 155648 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/fetch_suggestions.db  28672 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/frosting.db   212992 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/group_install.db      20480 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/install_queue.db      77824 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/install_source.db     32768 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/keyed_app_states.db   49152 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/library.db    49152 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/localappstate.db      376832 bytes
```

```
working/extracted/Pixel 3/data/data/com.android.vending/databases/localappstate.db      376832 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/offline_acquire_store.db      24576 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/phenotype.db  995328 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/prefetch_recommendation_store.db      86016 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/scheduled_acquisitions.db     24576 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/suggestions.db        20480 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/verify_apps.db        294912 bytes
working/extracted/Pixel 3/data/data/com.android.vending/databases/xternal_referrer_status.db    24576 bytes
working/extracted/Pixel 3/data/data/com.discord/databases/google_analytics_v4.db        28672 bytes
working/extracted/Pixel 3/data/data/com.discord/databases/google_app_measurement_local.db       16384 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/ads.db        28672 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/EmogiSDK.db   3936256 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/google_app_measurement_local.db       16384 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/__leanplum.db 36864 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/supersonic_sdk.db     16384 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/textnow_data.db       155648 bytes
working/extracted/Pixel 3/data/data/com.facebook.orca/cache/graph_store_cache/100046799400843/GraphStore.sqlite3        16384 bytes
working/extracted/Pixel 3/data/data/com.facebook.orca/databases/omnistore_100046799400843_v01.db        516096 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.chromecast.app/databases/-1_threads.notifications.db        20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.chromecast.app/databases/1_threads.notifications.db 20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.chromecast.app/databases/accounts.notifications.db  20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.chromecast.app/databases/google_app_measurement_local.db    16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.chromecast.app/databases/google_tagmanager.db       20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.chromecast.app/databases/growthkit.db       94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs/app_cello/thisisdfir@gmail.com/cello.db        4096 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs/databases/DocList.db   315392 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs/databases/google_analytics_v4.db       28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs/databases/growthkit.db 102400 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs.editors.docs/databases/DocList.db      512000 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs.editors.docs/databases/Editors.db      24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs.editors.docs/databases/google_analytics_v4.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs.editors.docs/databases/Storage.db      36864 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.dreamliner/databases/dreamliner.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.magazines/databases/google_app_measurement_local.db 16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_myplaces.db      40960 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_storage.db       94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_sync.db  45056 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/inbox_notifications.db       20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/ue3.db       24576 bytes
```

```
working/extracted/Pixel 3/data/data/com.google.android.apps.chromecast.app/databases/google_tagmanager.db       20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.chromecast.app/databases/growthkit.db       94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs/app_cello/thisisdfir@gmail.com/cello.db        4096 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs/databases/DocList.db   315392 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs/databases/google_analytics_v4.db       28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs/databases/growthkit.db 102400 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs.editors.docs/databases/DocList.db      512000 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs.editors.docs/databases/Editors.db      24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs.editors.docs/databases/google_analytics_v4.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.docs.editors.docs/databases/Storage.db      36864 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.dreamliner/databases/dreamliner.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.magazines/databases/google_app_measurement_local.db 16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_myplaces.db      40960 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_storage.db       94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_sync.db  45056 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/inbox_notifications.db       20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/ue3.db       24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/ugc_photos_location_data.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/files/offlinedatabase/offline.db       389120 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.messaging/databases/google_app_measurement_local.db 16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.messaging/databases/growthkit.db    102400 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.messaging/databases/imsft.db        24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.nexuslauncher/databases/app_icons.db        950272 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.nexuslauncher/databases/launcher.db 425984 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.nexuslauncher/databases/widgetpreviews.db   16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/google_app_measurement_local.db    16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/gphotos0.db        2191360 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/gphotos-1.db       1232896 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/local_trash.db     28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.safetyhub/files/accounts/1/SqliteKeyValueCache&#58;EmergencyContactsCache.db        24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.safetyhub/files/accounts/1/SqliteKeyValueCache&#58;MedInfoCache.db  24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.tachyon/app_crypto/crypto.db        53248 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.tachyon/databases/drishti.db        20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.tachyon/databases/growthkit.db      94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.tachyon/databases/tachyon.db        208896 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.tachyon/databases/TachyonGroupKeyValue.db   20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.tips/databases/google_app_measurement_local.db      16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.turbo/databases/appusage.db 45056 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.turbo/databases/turbo.db    81920 bytes
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "===== GLOBAL SETTINGS: SELECTED KEYS ====="
grep -iE 'wifi|bluetooth|airplane|device_name|adb_enabled|development_settings' \
  "working/extracted/Pixel 3/data/system/users/0/settings_global.xml" \
  | tee outputs/settings_global_selected.txt

echo
echo "===== SECURE SETTINGS: SELECTED KEYS ====="
grep -iE 'wifi|bluetooth|lockscreen|user_setup|location|android_id' \
  "working/extracted/Pixel 3/data/system/users/0/settings_secure.xml" \
  | tee outputs/settings_secure_selected.txt

echo
echo "===== SYSTEM SETTINGS: SELECTED KEYS ====="
grep -iE 'sound|screen_brightness|screen_off_timeout|ringtone' \
  "working/extracted/Pixel 3/data/system/users/0/settings_system.xml" \
  | tee outputs/settings_system_selected.txt
===== GLOBAL SETTINGS: SELECTED KEYS =====
  <setting id="199" name="development_settings_enabled" value="1" package="com.android.settings" defaultValue="1" defaultSysSet="true" />
  <setting id="23" name="wifi_display_on" value="0" package="android" defaultValue="0" defaultSysSet="true" />
  <setting id="46" name="device_name" value="Pixel 3" package="android" defaultValue="Pixel 3" defaultSysSet="true" />
  <setting id="14" name="wifi_networks_available_notification_on" value="1" package="android" defaultValue="1" defaultSysSet="true" />
  <setting id="5497" name="bluetooth_on" value="0" package="android" defaultValue="0" defaultSysSet="true" />
  <setting id="221" name="adb_enabled" value="1" package="android" defaultValue="1" defaultSysSet="true" />
  <setting id="2" name="airplane_mode_radios" value="cell,bluetooth,wifi,nfc,wimax" package="android" defaultValue="cell,bluetooth,wifi,nfc,wimax" defaultSysSet="
true" />
  <setting id="22" name="wifi_max_dhcp_retry_count" value="9" package="android" defaultValue="9" defaultSysSet="true" />
  <setting id="3" name="airplane_mode_toggleable_radios" value="bluetooth,wifi,nfc" package="android" defaultValue="bluetooth,wifi,nfc" defaultSysSet="true" />
  <setting id="50" name="bluetooth_disabled_profiles" value="0" package="android" defaultValue="0" defaultSysSet="true" />
  <setting id="8" name="wifi_sleep_policy" value="2" package="android" defaultValue="2" defaultSysSet="true" />
  <setting id="5816" name="wifi_scan_always_enabled" value="0" package="com.google.android.gms" defaultValue="0" defaultSysSet="true" />
  <setting id="0" name="airplane_mode_on" value="0" package="android" defaultValue="0" defaultSysSet="true" />
  <setting id="51" name="wifi_wakeup_enabled" value="1" package="android" defaultValue="1" defaultSysSet="true" />
  <setting id="5822" name="wifi_on" value="1" package="android" defaultValue="1" defaultSysSet="true" />

===== SECURE SETTINGS: SELECTED KEYS =====
```

```
=== SECURE SETTINGS: SELECTED KEYS ===
  <setting id="150" name="allowed_geolocation_origins" value="http://www.google.com http://www.google.co.uk" package="com.google.android.setupwizard" defaultValue
="http://www.google.com http://www.google.co.uk" defaultSysSet="true" />
  <setting id="146" name="location_mode" value="3" package="com.google.android.gms" defaultValue="3" defaultSysSet="true" />
  <setting id="129" name="masterLocationPackagePrefixBlacklist" value="com.google.,com.semaphoremobile.zagat.android" package="com.google.android.gsf" />
  <setting id="397" name="location_providers_allowed" value="gps,network" package="android" defaultValue="gps,network" defaultSysSet="true" />
  <setting id="153" name="user_setup_complete" value="1" package="com.google.android.setupwizard" defaultValue="1" defaultSysSet="true" />
  <setting id="10" name="lockscreen.disabled" value="0" package="android" defaultValue="0" defaultSysSet="true" />
  <setting id="131" name="masterLocationPackagePrefixWhitelist" value="com.google.android.gms" package="com.google.android.gsf" />
  <setting id="126" name="lockscreen.options" value="enable_facelock" package="com.google.android.gsf" />
  <setting id="0" name="mock_location" value="0" package="android" defaultValue="0" defaultSysSet="true" />
  <setting id="65" name="bluetooth_name" value="Pixel 3" package="android" defaultValue="Pixel 3" defaultSysSet="true" />
  <setting id="25" name="android_id" value="811b7f444043106a" package="android" defaultValue="811b7f444043106a" defaultSysSet="true" />
  <setting id="111" name="sysui_qs_tiles" value="wifi,bt,dnd,flashlight,rotation,battery,cell,airplane,night" package="com.android.systemui" defaultValue="wifi,bt
,dnd,flashlight,rotation,battery,cell,airplane,night" defaultSysSet="true" />
  <setting id="71" name="bluetooth_address" value="7C:D9:5C:AC:A2:CE" package="android" defaultValue="7C:D9:5C:AC:A2:CE" defaultSysSet="true" />
  <setting id="152" name="user_setup_personalization_state" value="10" package="com.google.android.setupwizard" defaultValue="10" defaultSysSet="true" />

=== SYSTEM SETTINGS: SELECTED KEYS ===
  <setting id="39" name="notification_sound" value="content://media/internal/audio/media/39?title=Popcorn&amp;canonical=1" package="com.android.providers.media" /
>
  <setting id="37" name="ringtone" value="content://media/internal/audio/media/76?title=The%20Big%20Adventure&amp;canonical=1" package="com.android.providers.medi
a" />
  <setting id="16" name="screen_brightness_mode" value="1" package="android" defaultValue="1" defaultSysSet="true" />
  <setting id="15" name="screen_brightness_for_vr" value="111" package="android" defaultValue="111" defaultSysSet="true" />
  <setting id="225" name="ringtone_set" value="1" package="com.google.android.gsf" />
  <setting id="223" name="notification_sound_set" value="1" package="com.google.android.gsf" />
  <setting id="22" name="lockscreen_sounds_enabled" value="1" package="android" defaultValue="1" defaultSysSet="true" />
  <setting id="55" name="screen_off_timeout" value="300000" package="com.android.settings" defaultValue="300000" defaultSysSet="true" />
  <setting id="729" name="sound_effects_enabled" value="1" package="com.google.android.gms" defaultValue="1" defaultSysSet="true" />
  <setting id="877" name="screen_brightness" value="101" package="android" defaultValue="101" defaultSysSet="true" />

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ grep -iE 'mmssms|telephony|contacts|calllog|calendar|events|gservices|messages|dialer|whatsapp|textnow|kik|location|maps|photos' \
  outputs/database_inventory.txt \
  | tee outputs/database_candidates.txt
working/extracted/Pixel 3/data/data/com.android.providers.calendar/databases/calendar.db        122880 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db 40960 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db       376832 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/profile.db 376832 bytes
working/extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db 118784 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/ads.db        28672 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/EmogiSDK.db   3936256 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/google_app_measurement_local.db       16384 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/__leanplum.db 36864 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/supersonic_sdk.db     16384 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/textnow_data.db       155648 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_myplaces.db      40960 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_storage.db       94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_sync.db  45056 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/inbox_notifications.db       20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/ue3.db       24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/ugc_photos_location_data.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/files/offlinedatabase/offline.db       389120 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/google_app_measurement_local.db    16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/gphotos0.db        2191360 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/gphotos-1.db       1232896 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/local_trash.db     28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.safetyhub/files/accounts/1/SqliteKeyValueCache6#58;EmergencyContactsCache.db        24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.as/databases/reflection_gel_events.db    696320 bytes
working/extracted/Pixel 3/data/data/com.google.android.calendar/databases/google_analytics_v2.db        20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.calendar/databases/groovedata.db 28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.calendar/databases/timelydata.db 69632 bytes
working/extracted/Pixel 3/data/data/com.google.android.calendar/databases/usernotifications.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.contacts/databases/contacts.db   102400 bytes
working/extracted/Pixel 3/data/data/com.google.android.contacts/databases/growthkit.db  94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.dialer/databases/annotated_call_log.db   24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.dialer/databases/dialer.db       77824 bytes
working/extracted/Pixel 3/data/data/com.google.android.dialer/databases/growthkit.db    94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.dialer/databases/normalized_spam.db      7348224 bytes
```

```
┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF205-Case1

mkdir -p working/databases

cp -p \
  "working/extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db" \
  working/databases/mmssms.db

sha256sum \
  "working/extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db" \
  working/databases/mmssms.db \
  | tee outputs/mmssms_copy_hashes.txt
cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342  working/extracted/Pixel 3/data/data/com.android.providers.telephony/databas
es/mmssms.db
cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342  working/databases/mmssms.db
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/mmssms_counts.txt
.headers on
.mode column

SELECT 'sms' AS table_name, COUNT(*) AS record_count FROM sms;
SELECT 'pdu' AS table_name, COUNT(*) AS record_count FROM pdu;
SELECT 'part' AS table_name, COUNT(*) AS record_count FROM part;
SELECT 'addr' AS table_name, COUNT(*) AS record_count FROM addr;
SQL
table_name  record_count
──────────  ────────────
sms         24
table_name  record_count
──────────  ────────────
pdu         7
table_name  record_count
──────────  ────────────
part        10
table_name  record_count
──────────  ────────────
addr        14

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/mmssms_schema.txt
PRAGMA integrity_check;
.tables
.schema sms
.schema pdu
.schema part
.schema addr
SQL
ok
addr                pdu_restricted      threads
android_metadata    pending_msgs        words
attachments         rate                words_content
canonical_addresses raw                 words_segdir
drm                 sms                 words_segments
part                sms_restricted
pdu                 sr_pending
CREATE TABLE sms (_id INTEGER PRIMARY KEY,thread_id INTEGER,address TEXT,person INTEGER,date INTEGER,date_sent INTEGER DEFAULT 0,protocol INT
EGER,read INTEGER DEFAULT 0,status INTEGER DEFAULT -1,type INTEGER,reply_path_present INTEGER,subject TEXT,body TEXT,service_center TEXT,lock
ed INTEGER DEFAULT 0,sub_id INTEGER DEFAULT -1, error_code INTEGER DEFAULT 0,creator TEXT,seen INTEGER DEFAULT 0);
CREATE TRIGGER sms_update_thread_on_insert AFTER INSERT ON sms BEGIN  UPDATE threads SET    date = (strftime('%s','now') * 1000),     snippet
 = new.body,     snippet_cs = 0  WHERE threads._id = new.thread_id;   UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms
 LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id        AND sms.type <> 3) +      (SELECT COUNT(pdu._
id) FROM pdu LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id        AND (m_type=132 OR m_type=130 OR
m_type=128)        AND msg_box <> 3)   WHERE threads._id = new.thread_id;   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM
 sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id
; END;
CREATE TRIGGER sms_update_thread_date_subject_on_update AFTER  UPDATE OF date, body, type  ON sms BEGIN  UPDATE threads SET    date = (strfti
me('%s','now') * 1000),     snippet = new.body,     snippet_cs = 0  WHERE threads._id = new.thread_id;   UPDATE threads SET message_count =
    (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id        AND sms.ty
pe <> 3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id
  AND (m_type=132 OR m_type=130 OR m_type=128)        AND msg_box <> 3)   WHERE threads._id = new.thread_id;   UPDATE threads SET read =
CASE (SELECT COUNT(*)          FROM sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END
  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER sms_update_thread_read_on_update AFTER  UPDATE OF read  ON sms BEGIN   UPDATE threads SET read =     CASE (SELECT COUNT(*)
      FROM sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new
.thread_id; END;
CREATE TRIGGER update_threads_error_on_update_sms   AFTER UPDATE OF type ON sms  WHEN (OLD.type <> 5 AND NEW.type = 5)    OR (OLD.type = 5 AN
```

```
Session  Actions  Edit  View  Help
m_type=128)        AND msg_box ≠ 3)   WHERE threads._id = new.thread_id;   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM
 sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id
; END;
CREATE TRIGGER sms_update_thread_date_subject_on_update AFTER  UPDATE OF date, body, type  ON sms BEGIN  UPDATE threads SET    date = (strfti
me('%s','now') * 1000),     snippet = new.body,     snippet_cs = 0  WHERE threads._id = new.thread_id;   UPDATE threads SET message_count =
    (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id       AND sms.ty
pe ≠ 3) +     (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads       ON threads._id = thread_id      WHERE thread_id = new.thread_id
  AND (m_type=132 OR m_type=130 OR m_type=128)        AND msg_box ≠ 3)   WHERE threads._id = new.thread_id;   UPDATE threads SET read =
CASE (SELECT COUNT(*)          FROM sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END
  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER sms_update_thread_read_on_update AFTER  UPDATE OF read  ON sms BEGIN   UPDATE threads SET read =     CASE (SELECT COUNT(*)
      FROM sms          WHERE read = 0            AND thread_id = threads._id)      WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new
.thread_id; END;
CREATE TRIGGER update_threads_error_on_update_sms   AFTER UPDATE OF type ON sms  WHEN (OLD.type ≠ 5 AND NEW.type = 5)    OR (OLD.type = 5 AN
D NEW.type ≠ 5) BEGIN   UPDATE threads SET error =     CASE      WHEN NEW.type = 5 THEN error + 1      ELSE error - 1    END   WHERE _id = N
EW.thread_id; END;
CREATE TRIGGER sms_words_update AFTER UPDATE ON sms BEGIN UPDATE words  SET index_text = NEW.body WHERE (source_id=NEW._id AND table_to_use=1
);  END;
CREATE TRIGGER sms_words_delete AFTER DELETE ON sms BEGIN DELETE FROM   words WHERE source_id = OLD._id AND table_to_use = 1; END;
CREATE INDEX typeThreadIdIndex ON sms (type, thread_id);
CREATE INDEX threadIdDateIndex ON sms (thread_id, date);
CREATE TABLE pdu (_id INTEGER PRIMARY KEY AUTOINCREMENT,thread_id INTEGER,date INTEGER,date_sent INTEGER DEFAULT 0,msg_box INTEGER,read INTEG
ER DEFAULT 0,m_id TEXT,sub TEXT,sub_cs INTEGER,ct_t TEXT,ct_l TEXT,exp INTEGER,m_cls TEXT,m_type INTEGER,v INTEGER,m_size INTEGER,pri INTEGER
,rr INTEGER,rpt_a INTEGER,resp_st INTEGER,st INTEGER,tr_id TEXT,retr_st INTEGER,retr_txt TEXT,retr_txt_cs INTEGER,read_status INTEGER,ct_cls
INTEGER,resp_txt TEXT,d_tm INTEGER,d_rpt INTEGER,locked INTEGER DEFAULT 0,sub_id INTEGER DEFAULT -1, seen INTEGER DEFAULT 0,creator TEXT,text
_only INTEGER DEFAULT 0);
CREATE TRIGGER part_cleanup DELETE ON pdu BEGIN   DELETE FROM part  WHERE mid=old._id;END;
CREATE TRIGGER addr_cleanup DELETE ON pdu BEGIN   DELETE FROM addr  WHERE msg_id=old._id;END;
CREATE TRIGGER cleanup_delivery_and_read_report AFTER DELETE ON pdu WHEN old.m_type=128 BEGIN   DELETE FROM pdu  WHERE (m_type=134    OR m_ty
pe=136)    AND m_id=old.m_id; END;
CREATE TRIGGER update_threads_on_update_pdu  AFTER UPDATE of thread_id ON pdu  BEGIN   UPDATE threads SET has_attachment=1 WHERE _id IN    (S
ELECT pdu.thread_id FROM part JOIN pdu      WHERE part.ct ≠ 'text/plain' AND part.ct ≠ 'application/smil'      AND part.mid = pdu._id); END
;
CREATE TRIGGER delete_mms_pending_on_delete AFTER DELETE ON pdu BEGIN   DELETE FROM pending_msgs  WHERE msg_id=old._id; END;
CREATE TRIGGER delete_mms_pending_on_update AFTER UPDATE ON pdu WHEN old.msg_box=4  AND new.msg_box≠4 BEGIN   DELETE FROM pending_msgs  WHER
E msg_id=new._id; END;
CREATE TRIGGER insert_mms_pending_on_insert AFTER INSERT ON pdu WHEN new.m_type=130  OR new.m_type=135 BEGIN   INSERT INTO pending_msgs    (p
roto_type,     msg_id,     msg_type,     err_type,     err_code,     retry_index,     due_time)   VALUES     (1,      new._id,      new.m_typ
e,0,0,0,0);END;
```

```
e,0,0,0,0);END;
CREATE TRIGGER insert_mms_pending_on_update AFTER UPDATE ON pdu WHEN new.m_type=128  AND new.msg_box=4  AND old.msg_box≠4 BEGIN   INSERT INT
O pending_msgs    (proto_type,     msg_id,     msg_type,     err_type,     err_code,     retry_index,     due_time)   VALUES     (1,      new
._id,      new.m_type,0,0,0,0);END;
CREATE TRIGGER pdu_update_thread_date_subject_on_update AFTER  UPDATE OF date, sub, msg_box  ON pdu   WHEN new.m_type=132    OR new.m_type=13
0    OR new.m_type=128 BEGIN  UPDATE threads SET    date = (strftime('%s','now') * 1000),     snippet = new.sub,     snippet_cs = new.sub_cs
 WHERE threads._id = new.thread_id;   UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON thre
ads._id = thread_id      WHERE thread_id = new.thread_id        AND sms.type ≠ 3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads
      ON threads._id = thread_id      WHERE thread_id = new.thread_id       AND (m_type=132 OR m_type=130 OR m_type=128)       AND msg_box !
= 3)   WHERE threads._id = new.thread_id;   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM pdu          WHERE read = 0
       AND thread_id = threads._id            AND (m_type=132 OR m_type=130 OR m_type=128))       WHEN 0 THEN 1      ELSE 0    END  WHERE t
hreads._id = new.thread_id; END;
CREATE TRIGGER pdu_update_thread_on_delete AFTER DELETE ON pdu BEGIN   UPDATE threads SET      date = (strftime('%s','now') * 1000)  WHERE th
reads._id = old.thread_id;   UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON threads._id =
 thread_id      WHERE thread_id = old.thread_id        AND sms.type ≠ 3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads       ON t
hreads._id = thread_id      WHERE thread_id = old.thread_id        AND (m_type=132 OR m_type=130 OR m_type=128)       AND msg_box ≠ 3)   WH
ERE threads._id = old.thread_id;   UPDATE threads SET snippet =    (SELECT snippet FROM     (SELECT date * 1000 AS date, sub AS snippet, thre
ad_id FROM pdu      UNION SELECT date, body AS snippet, thread_id FROM sms)    WHERE thread_id = OLD.thread_id ORDER BY date DESC LIMIT 1)
WHERE threads._id = OLD.thread_id;   UPDATE threads SET snippet_cs =    (SELECT snippet_cs FROM     (SELECT date * 1000 AS date, sub_cs AS sn
ippet_cs, thread_id FROM pdu      UNION SELECT date, 0 AS snippet_cs, thread_id FROM sms)    WHERE thread_id = OLD.thread_id ORDER BY date DE
SC LIMIT 1)   WHERE threads._id = OLD.thread_id; END;
CREATE TRIGGER pdu_update_thread_on_insert AFTER INSERT ON pdu   WHEN new.m_type=132    OR new.m_type=130    OR new.m_type=128 BEGIN  UPDATE
threads SET    date = (strftime('%s','now') * 1000),     snippet = new.sub,     snippet_cs = new.sub_cs  WHERE threads._id = new.thread_id;
 UPDATE threads SET message_count =      (SELECT COUNT(sms._id) FROM sms LEFT JOIN threads       ON threads._id = thread_id      WHERE thread
_id = new.thread_id       AND sms.type ≠ 3) +      (SELECT COUNT(pdu._id) FROM pdu LEFT JOIN threads       ON threads._id = thread_id
WHERE thread_id = new.thread_id        AND (m_type=132 OR m_type=130 OR m_type=128)       AND msg_box ≠ 3)   WHERE threads._id = new.thread
_id;   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM pdu          WHERE read = 0           AND thread_id = threads._id
          AND (m_type=132 OR m_type=130 OR m_type=128))       WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER pdu_update_thread_read_on_update AFTER  UPDATE OF read  ON pdu   WHEN new.m_type=132    OR new.m_type=130    OR new.m_type=128
 BEGIN   UPDATE threads SET read =     CASE (SELECT COUNT(*)          FROM pdu          WHERE read = 0           AND thread_id = threads._id
            AND (m_type=132 OR m_type=130 OR m_type=128))       WHEN 0 THEN 1      ELSE 0    END  WHERE threads._id = new.thread_id; END;
CREATE TRIGGER update_threads_error_on_delete_mms   BEFORE DELETE ON pdu  WHEN OLD._id IN (SELECT DISTINCT msg_id                    FROM pend
ing_msgs                   WHERE err_type ≥ 10) BEGIN   UPDATE threads SET error = error - 1  WHERE _id = OLD.thread_id; END;
CREATE TRIGGER update_threads_error_on_move_mms   BEFORE UPDATE OF msg_box ON pdu   WHEN (OLD.msg_box = 4 AND NEW.msg_box ≠ 4)   AND (OLD._i
d IN (SELECT DISTINCT msg_id                FROM pending_msgs                WHERE err_type ≥ 10)) BEGIN   UPDATE threads SET error =
error - 1  WHERE _id = OLD.thread_id; END;
CREATE TABLE part (_id INTEGER PRIMARY KEY AUTOINCREMENT,mid INTEGER,seq INTEGER DEFAULT 0,ct TEXT,name TEXT,chset INTEGER,cd TEXT,fn TEXT,ci
d TEXT,cl TEXT,ctt_s INTEGER,ctt_t TEXT,_data TEXT,text TEXT);
CREATE TRIGGER update_threads_on_insert_part  AFTER INSERT ON part  WHEN new.ct ≠ 'text/plain' AND new.ct ≠ 'application/smil'  BEGIN   UPD
```

```
CREATE TRIGGER update_threads_on_insert_part  AFTER INSERT ON part  WHEN new.ct ≠ 'text/plain' AND new.ct ≠ 'application/smil'  BEGIN   UPD
ATE threads SET has_attachment=1 WHERE _id IN    (SELECT pdu.thread_id FROM part JOIN pdu ON pdu._id=part.mid      WHERE part._id=new._id LIM
IT 1);  END;
CREATE TRIGGER update_threads_on_update_part  AFTER UPDATE of mid ON part  WHEN new.ct ≠ 'text/plain' AND new.ct ≠ 'application/smil'  BEGI
N   UPDATE threads SET has_attachment=1 WHERE _id IN    (SELECT pdu.thread_id FROM part JOIN pdu ON pdu._id=part.mid      WHERE part._id=new.
_id LIMIT 1);  END;
CREATE TRIGGER update_threads_on_delete_part  AFTER DELETE ON part  WHEN old.ct ≠ 'text/plain' AND old.ct ≠ 'application/smil'  BEGIN   UPD
ATE threads SET has_attachment =    CASE     (SELECT COUNT(*) FROM part JOIN pdu      WHERE pdu.thread_id = threads._id      AND part.ct ≠ '
text/plain' AND part.ct ≠ 'application/smil'      AND part.mid = pdu._id)   WHEN 0 THEN 0    ELSE 1    END;  END;
CREATE TRIGGER mms_words_update AFTER UPDATE ON part BEGIN UPDATE words  SET index_text = NEW.text WHERE (source_id=NEW._id AND table_to_use=
2);  END;
CREATE TRIGGER mms_words_delete AFTER DELETE ON part BEGIN DELETE FROM  words WHERE source_id = OLD._id AND table_to_use = 2; END;
CREATE INDEX partMidIndex ON part (mid);
CREATE TABLE addr (_id INTEGER PRIMARY KEY,msg_id INTEGER,contact_id INTEGER,address TEXT,type INTEGER,charset INTEGER);
CREATE INDEX addrMsgIdIndex ON addr (msg_id);
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/mmssms_counts.txt
.headers on
.mode column

SELECT 'sms' AS table_name, COUNT(*) AS record_count FROM sms;
SELECT 'pdu' AS table_name, COUNT(*) AS record_count FROM pdu;
SELECT 'part' AS table_name, COUNT(*) AS record_count FROM part;
SELECT 'addr' AS table_name, COUNT(*) AS record_count FROM addr;
SQL
table_name  record_count
──────────  ────────────
sms         24
table_name  record_count
──────────  ────────────
pdu         7
table_name  record_count
──────────  ────────────
part        10
table_name  record_count
──────────  ────────────
addr        14

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
20   13          1581185821854   2020-02-08 18:17:01  1     0     82
21   17          1581273702392   2020-02-09 18:41:42  2     1     56
22   13          1581444160908   2020-02-11 18:02:40  1     0     84
23   13          1581534738811   2020-02-12 19:12:18  1     0     83
24   13          1581628344576   2020-02-13 21:12:24  1     0     84
25   17          1581685743277   2020-02-14 13:09:03  2     1     46

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/sms_summary.txt
.headers on
.mode column
SELECT
    _id,
    thread_id,
    date,
    datetime(date/1000, 'unixepoch') AS date_utc,
    type,
    read,
    length(body) AS body_length
FROM sms
ORDER BY date;
SQL
```

_id	thread_id	date	date_utc	type	read	body_length
2	4	1580322045251	2020-01-29 18:20:45	1	0	70
3	5	1580322646631	2020-01-29 18:30:46	1	0	87
4	6	1580322696889	2020-01-29 18:31:36	1	0	55
5	7	1580322728551	2020-01-29 18:32:08	1	0	37
6	8	1580322820355	2020-01-29 18:33:40	1	0	37
7	9	1580322903320	2020-01-29 18:35:03	1	0	37
8	10	1580323028883	2020-01-29 18:37:08	1	0	153
9	5	1580323298115	2020-01-29 18:41:38	1	0	43
10	11	1580323388596	2020-01-29 18:43:08	1	0	54
11	12	1580323813826	2020-01-29 18:50:13	1	0	47
12	13	1580323824938	2020-01-29 18:50:24	1	0	194
13	5	1580328970337	2020-01-29 20:16:10	1	0	55
14	14	1580332533357	2020-01-29 21:15:33	1	0	66
15	15	1580518803531	2020-02-01 01:00:03	1	1	42
16	16	1580519052798	2020-02-01 01:04:12	1	1	88
17	13	1580757526393	2020-02-03 19:18:46	1	0	86
18	13	1580919741561	2020-02-05 16:22:21	1	0	82
19	13	1581063359840	2020-02-07 08:15:59	1	0	83
20	13	1581185821854	2020-02-08 18:17:01	1	0	82

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

sqlite3 working/databases/mmssms.db <<'SQL' | tee outputs/sms_details.txt
.headers on
.mode column
.width 5 8 22 8 5 6 70
SELECT
    _id,
    thread_id,
    address,
    datetime(date/1000, 'unixepoch') AS date_utc,
    type,
    read,
    body
FROM sms
ORDER BY date;
SQL
_id    thread_i  address                 date_utc  type   read    body
-----  --------  ----------------------  --------  -----  ------  ----------------------------------------------------------------------
2      4         +13233205525            2020-01-  1      0       <#> Please enter 304868 into LINE within the next 30 mins.
                                         29 18:20                 JFoQLtyexga
                                         :45

3      5         732873                  2020-01-  1      0       Telegram code: 59389
                                         29 18:30
                                         :46                      You can also tap on this link to log in:
                                                                  https://t.me/login/59389

4      6         +18437514985            2020-01-  1      0       <#> Your Signal verification code: 303-077
                                         29 18:31
                                         :36                      doDiFGKPO1r

5      7         +12253690855            2020-01-  1      0       Your imo code is 685614
                                         29 18:32                 9K+9OFePKE9
                                         :08

6      8         +12347570651            2020-01-  1      0       Your Skout verification code is 2661.
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cat outputs/profile_login_times_utc.txt
cat outputs/database_candidates.txt
User ID 0: 2020-02-14T15:51:57.124000+00:00
User ID 11: 2020-02-14T12:50:01.922000+00:00
working/extracted/Pixel 3/data/data/com.android.providers.calendar/databases/calendar.db        122880 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db 40960 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/contacts2.db       376832 bytes
working/extracted/Pixel 3/data/data/com.android.providers.contacts/databases/profile.db 376832 bytes
working/extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db 118784 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/ads.db        28672 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/EmogiSDK.db   3936256 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/google_app_measurement_local.db       16384 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/__leanplum.db 36864 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/supersonic_sdk.db     16384 bytes
working/extracted/Pixel 3/data/data/com.enflick.android.TextNow/databases/textnow_data.db       155648 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_myplaces.db      40960 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_storage.db       94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/gmm_sync.db  45056 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/inbox_notifications.db       20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/ue3.db       24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/databases/ugc_photos_location_data.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.maps/files/offlinedatabase/offline.db       389120 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/google_app_measurement_local.db    16384 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/gphotos0.db        2191360 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/gphotos-1.db       1232896 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.photos/databases/local_trash.db     28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.apps.safetyhub/files/accounts/1/SqliteKeyValueCache&#58;EmergencyContactsCache.db        24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.as/databases/reflection_gel_events.db    696320 bytes
working/extracted/Pixel 3/data/data/com.google.android.calendar/databases/google_analytics_v2.db        20480 bytes
working/extracted/Pixel 3/data/data/com.google.android.calendar/databases/groovedata.db 28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.calendar/databases/timelydata.db 69632 bytes
working/extracted/Pixel 3/data/data/com.google.android.calendar/databases/usernotifications.db  28672 bytes
working/extracted/Pixel 3/data/data/com.google.android.contacts/databases/contacts.db   102400 bytes
working/extracted/Pixel 3/data/data/com.google.android.contacts/databases/growthkit.db  94208 bytes
working/extracted/Pixel 3/data/data/com.google.android.dialer/databases/annotated_call_log.db   24576 bytes
working/extracted/Pixel 3/data/data/com.google.android.dialer/databases/dialer.db       77824 bytes
```

```
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ cd ~

mkdir -p ~/SBT-DF205-Case1/{original,working,evidence,outputs,notes,report,screenshots}

cd ~/SBT-DF205-Case1

pwd
/home/kali/SBT-DF205-Case1

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ find . -maxdepth 1 -type d | sort
.
./evidence
./notes
./original
./outputs
./report
./screenshots
./working

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "===== UTC TIME ====="
date -u '+%Y-%m-%d %H:%M:%S UTC'

echo
echo "===== KALI VERSION ====="
cat /etc/os-release | grep -E '^(PRETTY_NAME|VERSION)='

echo
echo "===== PYTHON ====="
python3 --version

echo
echo "===== SQLITE ====="
sqlite3 --version

echo
echo "===== SHA256SUM ====="
sha256sum --version | head -n 1

echo
echo "===== UNZIP ====="
unzip -v | head -n 2

echo
echo "===== EXIFTOOL ====="
exiftool -ver 2>&1
===== UTC TIME =====
2026-10-26 13:49:37 UTC

===== KALI VERSION =====
PRETTY_NAME="Kali GNU/Linux Rolling"
VERSION="2025.3"

===== PYTHON =====
```

```
===== PYTHON =====
Python 3.14.7

===== SQLITE =====
3.46.1 2024-08-13 09:16:08 c9c2ab54ba1f5f46360f1b4f35d849cd3f080e6fc2b6c60e91b16c63f69aalt1 (64-bit)

===== SHA256SUM =====
sha256sum (GNU coreutils) 9.7

===== UNZIP =====
UnZip 6.00 of 20 April 2009, by Debian. Original by Info-ZIP.


===== EXIFTOOL =====
13.25

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

{
    echo "Case Reference: SBT-DF205-C1"
    echo "Examiner: Charles Uke"
    echo "Recorded UTC: $(date -u '+%Y-%m-%d %H:%M:%S UTC')"
    echo
    echo "===== SYSTEM ====="
    cat /etc/os-release
    echo
    echo "===== TOOL VERSIONS ====="
    python3 --version 2>&1
    sqlite3 --version 2>&1
    sha256sum --version 2>&1 | head -n 1
    unzip -v 2>&1 | head -n 2
    exiftool -ver 2>&1
} | tee notes/initial_environment.txt
Case Reference: SBT-DF205-C1
Examiner: Charles Uke
Recorded UTC: 2026-10-26 13:50:36 UTC
```

```
cp --preserve=timestamps \
  "/home/kali/Downloads/Pixel 3.zip" \
  "original/Pixel 3.zip"

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sha256sum \
  "/home/kali/Downloads/Pixel 3.zip" \
  "original/Pixel 3.zip" \
  | tee notes/archive_integrity_check.txt
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  /home/kali/Downloads/Pixel 3.zip
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  original/Pixel 3.zip

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ stat --printf='%n: %s bytes\n' \
  "/home/kali/Downloads/Pixel 3.zip" \
  "original/Pixel 3.zip"
/home/kali/Downloads/Pixel 3.zip: 5247820897 bytes
original/Pixel 3.zip: 5247820897 bytes

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

cat > notes/Handling_Log.csv <<'EOF'
CaseReference,DateTimeUTC,Examiner,Action,SourcePath,DestinationPath,Tool,Result,Notes
SBT-DF205-C1,2026-10-26 13:49:37 UTC,Charles Uke,Environment check,/home/kali,,Kali GNU/Linux Rolling 2025.3,Recorded,"Initial environment time from terminal; actual archive receipt time not yet established"
EOF

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

stat --printf='Filename: %n\nSize: %s bytes\nModified: %y\n' \
  "/home/kali/Downloads/Pixel 3.zip"

sha256sum "/home/kali/Downloads/Pixel 3.zip" \
  | tee notes/archive_source_sha256.txt
Filename: /home/kali/Downloads/Pixel 3.zip
Size: 5247820897 bytes
Modified: 2026-10-26 07:43:00.749374382 -0400
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  /home/kali/Downloads/Pixel 3.zip

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

cp --preserve=timestamps \
  "/home/kali/Downloads/Pixel 3.zip" \
  "original/Pixel 3.zip"

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ sha256sum \
  "/home/kali/Downloads/Pixel 3.zip" \
  "original/Pixel 3.zip" \
  | tee notes/archive_integrity_check.txt
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  /home/kali/Downloads/Pixel 3.zip
ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a  original/Pixel 3.zip

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ stat --printf='%n: %s bytes\n' \
  "/home/kali/Downloads/Pixel 3.zip" \
  "original/Pixel 3.zip"
/home/kali/Downloads/Pixel 3.zip: 5247820897 bytes
original/Pixel 3.zip: 5247820897 bytes

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "═══ ZIP INTEGRITY TEST SUMMARY ═══"
tail -n 3 outputs/archive_test.txt

echo
echo "═══ FIRST 40 ARCHIVE PATHS ═══"
head -n 40 outputs/archive_listing.txt
═══ ZIP INTEGRITY TEST SUMMARY ═══
    testing: Pixel 3/vendor/usr/keylayout/   OK
    testing: Pixel 3/vendor/usr/keylayout/uinput-fpc.kl   OK
No errors detected in compressed data of original/Pixel 3.zip.

═══ FIRST 40 ARCHIVE PATHS ═══
Archive:  original/Pixel 3.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  2020-02-15 11:08   Pixel 3/
    10244  2020-02-15 10:59   Pixel 3/.DS_Store
        0  2020-02-15 11:04   Pixel 3/apex/
        0  2020-02-15 11:00   Pixel 3/apex/com.android.apex.cts.shim/
        0  2020-02-15 11:00   Pixel 3/apex/com.android.apex.cts.shim@1/
       58  1980-01-01 00:00   Pixel 3/apex/com.android.apex.cts.shim@1/apex_manifest.json
        0  2020-02-15 10:45   Pixel 3/apex/com.android.apex.cts.shim@1/etc/
      774  1980-01-01 00:00   Pixel 3/apex/com.android.apex.cts.shim@1/etc/hash.txt
       58  1980-01-01 00:00   Pixel 3/apex/com.android.apex.cts.shim/apex_manifest.json
        0  2020-02-15 10:45   Pixel 3/apex/com.android.apex.cts.shim/etc/
      774  1980-01-01 00:00   Pixel 3/apex/com.android.apex.cts.shim/etc/hash.txt
        0  2020-02-15 11:00   Pixel 3/apex/com.android.conscrypt/
        0  2020-02-15 11:00   Pixel 3/apex/com.android.conscrypt@291900700/
       62  1980-01-01 00:00   Pixel 3/apex/com.android.conscrypt@291900700/apex_manifest.json
       62  1980-01-01 00:00   Pixel 3/apex/com.android.conscrypt/apex_manifest.json
        0  2020-02-15 11:00   Pixel 3/apex/com.android.media/
        0  2020-02-15 11:04   Pixel 3/apex/com.android.media.swcodec/
        0  2020-02-15 11:04   Pixel 3/apex/com.android.media.swcodec@292000500/
       66  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec@292000500/apex_manifest.json
        0  2020-02-15 10:45   Pixel 3/apex/com.android.media.swcodec@292000500/bin/
```

```
Session  Actions  Edit  View  Help
       0  2020-02-15 11:04   Pixel 3/apex/com.android.media.swcodec/
       0  2020-02-15 11:04   Pixel 3/apex/com.android.media.swcodec@292000500/
      66  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec@292000500/apex_manifest.json
       0  2020-02-15 10:45   Pixel 3/apex/com.android.media.swcodec@292000500/bin/
   16024  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec@292000500/bin/mediaswcodec
       0  2020-02-15 11:04   Pixel 3/apex/com.android.media.swcodec@292000500/etc/
     210  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec@292000500/etc/init.rc
    7216  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec@292000500/etc/ld.config.txt
   17455  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec@292000500/etc/media_codecs.xml
       0  2020-02-15 10:45   Pixel 3/apex/com.android.media.swcodec@292000500/etc/seccomp_policy/
    1869  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec@292000500/etc/seccomp_policy/mediaswcodec.policy
      66  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec/apex_manifest.json
       0  2020-02-15 10:45   Pixel 3/apex/com.android.media.swcodec/bin/
   16024  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec/bin/mediaswcodec
       0  2020-02-15 11:04   Pixel 3/apex/com.android.media.swcodec/etc/
     210  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec/etc/init.rc
    7216  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec/etc/ld.config.txt
   17455  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec/etc/media_codecs.xml
       0  2020-02-15 10:45   Pixel 3/apex/com.android.media.swcodec/etc/seccomp_policy/
    1869  1980-01-01 00:00   Pixel 3/apex/com.android.media.swcodec/etc/seccomp_policy/mediaswcodec.policy
       0  2020-02-15 11:00   Pixel 3/apex/com.android.media@292000300/

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ df -h ~
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        79G   42G   33G  56% /

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

mkdir -p working/extracted

echo "Extraction destination:"
realpath working/extracted
Extraction destination:
/home/kali/SBT-DF205-Case1/working/extracted

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

date -u '+Extraction start: %Y-%m-%d %H:%M:%S UTC' \
  | tee notes/extraction_times.txt

unzip -q "original/Pixel 3.zip" \
  -d working/extracted \
  > outputs/extraction.log 2>&1

status=$?

date -u '+Extraction end: %Y-%m-%d %H:%M:%S UTC' \
  | tee -a notes/extraction_times.txt

echo "Unzip exit status: $status"
echo "Extraction log: outputs/extraction.log"
Extraction start: 2026-10-26 15:10:17 UTC
zsh: read-only variable: status

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "===== EXTRACTED ROOT ====="
ls -ld working/extracted/"Pixel 3"

echo
echo "===== DIRECTORY SIZE ====="
du -sh working/extracted/"Pixel 3"

echo
echo "===== DISK SPACE AFTER EXTRACTION ====="
df -h ~

echo
echo "===== EXTRACTION LOG SUMMARY ====="
tail -n 10 outputs/extraction.log
===== EXTRACTED ROOT =====
```

```
echo "===== EXTRACTED ROOT ====="
ls -ld working/extracted/"Pixel 3"

echo
echo "===== DIRECTORY SIZE ====="
du -sh working/extracted/"Pixel 3"

echo
echo "===== DISK SPACE AFTER EXTRACTION ====="
df -h ~

echo
echo "===== EXTRACTION LOG SUMMARY ====="
tail -n 10 outputs/extraction.log
===== EXTRACTED ROOT =====
drwxrwxr-x 11 kali kali 4096 Feb 15  2020 'working/extracted/Pixel 3'

===== DIRECTORY SIZE =====
8.6G    working/extracted/Pixel 3

===== DISK SPACE AFTER EXTRACTION =====
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        79G   51G   25G  68% /

===== EXTRACTION LOG SUMMARY =====

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ find working/extracted/"Pixel 3" -maxdepth 2 -type d \
  \( -name data -o -name system -o -name vendor \) -print
working/extracted/Pixel 3/data
working/extracted/Pixel 3/data/data
working/extracted/Pixel 3/data/vendor
working/extracted/Pixel 3/data/system
working/extracted/Pixel 3/mnt/vendor
working/extracted/Pixel 3/vendor
working/extracted/Pixel 3/system

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

python3 - <<'PY' | tee outputs/profile_login_times_utc.txt
from datetime import datetime, timezone

profiles = {
    "User ID 0": 1581695517124,
    "User ID 11": 1581684601922,
}

for profile, milliseconds in profiles.items():
    dt = datetime.fromtimestamp(milliseconds / 1000, tz=timezone.utc)
    print(f"{profile}: {dt.isoformat()}")
PY
User ID 0: 2020-02-14T15:51:57.124000+00:00
User ID 11: 2020-02-14T12:50:01.922000+00:00
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "═══ BUILD.PROP FILES ═══"
find "working/extracted/Pixel 3" -type f -name build.prop -print
═══ BUILD.PROP FILES ═══
working/extracted/Pixel 3/product/build.prop
working/extracted/Pixel 3/sbin/.magisk/mirror/product/build.prop
working/extracted/Pixel 3/sbin/.magisk/mirror/vendor/build.prop
working/extracted/Pixel 3/sbin/.magisk/mirror/vendor/odm/etc/build.prop
working/extracted/Pixel 3/sbin/.magisk/mirror/system_root/system/build.prop
working/extracted/Pixel 3/vendor/build.prop
working/extracted/Pixel 3/vendor/odm/etc/build.prop
working/extracted/Pixel 3/system/build.prop

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ find "working/extracted/Pixel 3" -type f -name build.prop \
  -exec grep -H -E \
  '^(ro\.product\.manufacturer|ro\.product\.model|ro\.product\.brand|ro\.build\.version\.release|ro\.build\.id|ro\.build\.version\.security_patch|ro\.build\.finge
rprint)=' {} \; \
  | tee outputs/device_build_properties.txt
working/extracted/Pixel 3/sbin/.magisk/mirror/system_root/system/build.prop:ro.build.id=QQ1A.200105.003
working/extracted/Pixel 3/sbin/.magisk/mirror/system_root/system/build.prop:ro.build.version.release=10
working/extracted/Pixel 3/sbin/.magisk/mirror/system_root/system/build.prop:ro.build.version.security_patch=2020-01-01
working/extracted/Pixel 3/system/build.prop:ro.build.id=QQ1A.200105.003
working/extracted/Pixel 3/system/build.prop:ro.build.version.release=10
working/extracted/Pixel 3/system/build.prop:ro.build.version.security_patch=2020-01-01
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "═══ USER PROFILE DIRECTORY ═══"
ls -la "working/extracted/Pixel 3/data/system/users"

echo
echo "═══ USER PROFILE FILES ═══"
find "working/extracted/Pixel 3/data/system/users" \
  -maxdepth 2 -type f -printf '%p\t%s bytes\n' | sort
═══ USER PROFILE DIRECTORY ═══
total 28
drwxrwxr-x  4 kali kali 4096 Feb 15  2020 .
drwxrwxr-x 14 kali kali 4096 Feb 15  2020 ..
drwxrwxr-x  3 kali kali 4096 Feb 15  2020 0
-rw-rw-r--  1 kali kali  345 Feb 14  2020 0.xml
drwxrwxr-x  3 kali kali 4096 Feb 15  2020 11
-rw-rw-r--  1 kali kali  359 Feb 14  2020 11.xml
-rw-rw-r--  1 kali kali  356 Feb 13  2020 userlist.xml
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

cat > notes/device_profile_findings.txt <<'EOF'
SBT-DF205 Case Study 1 — Preliminary Device and Profile Findings

Android build properties:
- Android release: 10
- Build ID: QQ1A.200105.003
- Security patch level: 2020-01-01
- Source: working/extracted/Pixel 3/system/build.prop
- Note: Manufacturer, model, brand and fingerprint were not returned by the initial property search. Further corroboration is required.

Android user profiles:
- Source: working/extracted/Pixel 3/data/system/users/userlist.xml
- User ID 0 is listed.
- User ID 11 is listed.
- nextSerialNumber is 12.
- Guest restrictions include no_sms, no_install_unknown_sources, no_config_wifi and no_outgoing_calls.
- The XML alone does not establish the real-world identity of either profile's user.

Extraction:
- Extracted file count: 64550
- Verification observation recorded at 2026-10-26 15:28:14 UTC.
- Disk space available at verification: 25G.
- Original archive must remain unchanged.
EOF

cat notes/device_profile_findings.txt
SBT-DF205 Case Study 1 — Preliminary Device and Profile Findings

Android build properties:
- Android release: 10
- Build ID: QQ1A.200105.003
- Security patch level: 2020-01-01
- Source: working/extracted/Pixel 3/system/build.prop
- Note: Manufacturer, model, brand and fingerprint were not returned by the initial property search. Further corroboration is required.
```

```
- Disk space available at verification: 25G.
- Original archive must remain unchanged.
EOF

cat notes/device_profile_findings.txt
SBT-DF205 Case Study 1 — Preliminary Device and Profile Findings

Android build properties:
- Android release: 10
- Build ID: QQ1A.200105.003
- Security patch level: 2020-01-01
- Source: working/extracted/Pixel 3/system/build.prop
- Note: Manufacturer, model, brand and fingerprint were not returned by the initial property search. Further corroboration is required.

Android user profiles:
- Source: working/extracted/Pixel 3/data/system/users/userlist.xml
- User ID 0 is listed.
- User ID 11 is listed.
- nextSerialNumber is 12.
- Guest restrictions include no_sms, no_install_unknown_sources, no_config_wifi and no_outgoing_calls.
- The XML alone does not establish the real-world identity of either profile's user.

Extraction:
- Extracted file count: 64550
- Verification observation recorded at 2026-10-26 15:28:14 UTC.
- Disk space available at verification: 25G.
- Original archive must remain unchanged.
```

```
┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF205-Case1

mkdir -p working/databases

cp -p \
  "working/extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db" \
  working/databases/mmssms.db

sha256sum \
  "working/extracted/Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db" \
  working/databases/mmssms.db \
  | tee outputs/mmssms_copy_hashes.txt
cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342  working/extracted/Pixel 3/data/data/com.android.providers.telephony/databas
es/mmssms.db
cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342  working/databases/mmssms.db
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cat >> notes/device_profile_findings.txt <<'EOF'

Additional profile metadata:
- User ID 0 display name: This Is
- User ID 11 display name: User 2
- Both profile XML files contain the build fingerprint:
  google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
- User ID 0 lastLoggedIn value: 1581695517124 milliseconds since Unix epoch.
- User ID 11 lastLoggedIn value: 1581684601922 milliseconds since Unix epoch.
- Converted UTC values are recorded in outputs/profile_login_times_utc.txt.
- These are timestamps recorded in the image; their meaning depends on the Android artifact and should not be treated as proof of a particular person's actions.
EOF
```

```
═══ USER PROFILE FILES ═══
working/extracted/Pixel 3/data/system/users/0/app_idle_stats.xml        51425 bytes
working/extracted/Pixel 3/data/system/users/0/appwidgets.xml    76 bytes
working/extracted/Pixel 3/data/system/users/0/package-restrictions.xml  148541 bytes
working/extracted/Pixel 3/data/system/users/0/photo.png 4618 bytes
working/extracted/Pixel 3/data/system/users/0/roles.xml 915 bytes
working/extracted/Pixel 3/data/system/users/0/runtime-permissions.xml   74805 bytes
working/extracted/Pixel 3/data/system/users/0/settings_config.xml       33305 bytes
working/extracted/Pixel 3/data/system/users/0/settings_global.xml       25315 bytes
working/extracted/Pixel 3/data/system/users/0/settings_secure.xml       21631 bytes
working/extracted/Pixel 3/data/system/users/0/settings_ssaid.xml        5374 bytes
working/extracted/Pixel 3/data/system/users/0/settings_system.xml       5593 bytes
working/extracted/Pixel 3/data/system/users/0/wallpaper 3586142 bytes
working/extracted/Pixel 3/data/system/users/0/wallpaper_info.xml        282 bytes
working/extracted/Pixel 3/data/system/users/0/wallpaper_orig    3586142 bytes
working/extracted/Pixel 3/data/system/users/0.xml       345 bytes
working/extracted/Pixel 3/data/system/users/11/app_idle_stats.xml       51313 bytes
working/extracted/Pixel 3/data/system/users/11/appwidgets.xml   76 bytes
working/extracted/Pixel 3/data/system/users/11/device_policies.xml      136 bytes
working/extracted/Pixel 3/data/system/users/11/package-restrictions.xml 69154 bytes
working/extracted/Pixel 3/data/system/users/11/photo.png        4675 bytes
working/extracted/Pixel 3/data/system/users/11/roles.xml        915 bytes
working/extracted/Pixel 3/data/system/users/11/runtime-permissions.xml  56305 bytes
working/extracted/Pixel 3/data/system/users/11/settings_secure.xml      16542 bytes
working/extracted/Pixel 3/data/system/users/11/settings_ssaid.xml       1611 bytes
working/extracted/Pixel 3/data/system/users/11/settings_system.xml      4864 bytes
working/extracted/Pixel 3/data/system/users/11/wallpaper        1842674 bytes
working/extracted/Pixel 3/data/system/users/11/wallpaper_info.xml       276 bytes
working/extracted/Pixel 3/data/system/users/11/wallpaper_orig   1842674 bytes
working/extracted/Pixel 3/data/system/users/11.xml      359 bytes
working/extracted/Pixel 3/data/system/users/userlist.xml        356 bytes
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ if [ -f "working/extracted/Pixel 3/data/system/users/userlist.xml" ]; then
    cat "working/extracted/Pixel 3/data/system/users/userlist.xml"
else
    echo "userlist.xml not present at the expected path"
fi
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<users nextSerialNumber="12" version="7">
    <guestRestrictions>
        <restrictions no_sms="true" no_install_unknown_sources="true" no_config_wifi="true" no_outgoing_calls="true" />
    </guestRestrictions>
    <deviceOwnerUserId id="-10000" />
    <user id="0" />
    <user id="11" />
</users>

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

date -u '+Observed extraction verification: %Y-%m-%d %H:%M:%S UTC' \
  | tee -a notes/extraction_times.txt

echo "===== DISK SPACE ====="
df -h ~

echo
echo "===== EXTRACTED FILE COUNT ====="
find "working/extracted/Pixel 3" -type f | wc -l
Observed extraction verification: 2026-10-26 15:28:14 UTC
===== DISK SPACE =====
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        79G   51G   25G  68% /

===== EXTRACTED FILE COUNT =====
64550

┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case1]
└─$ cd ~/SBT-DF205-Case1

echo "═══ USER 0 PROFILE METADATA ═══"
cat "working/extracted/Pixel 3/data/system/users/0.xml"

echo
echo "═══ USER 11 PROFILE METADATA ═══"
cat "working/extracted/Pixel 3/data/system/users/11.xml"
═══ USER 0 PROFILE METADATA ═══
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="0" serialNumber="0" flags="19" created="0" lastLoggedIn="1581695517124" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:use
r/release-keys" icon="/data/system/users/0/photo.png" profileBadge="0">
    <name>This Is</name>
    <restrictions />
</user>

═══ USER 11 PROFILE METADATA ═══
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="1581684601922" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.0
03/6079943:user/release-keys" icon="/data/system/users/11/photo.png" profileBadge="0">
    <name>User 2</name>
    <restrictions />
</user>
```